



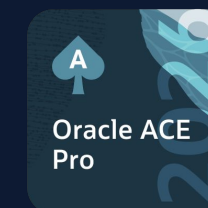
Oracle SQLcl: The Free Tool Every PL/SQL and APEX Developer Needs (Hey DBAs, you need it too)

Friday, Sep 04, 2026 at 12:40 AM - 13:25 PM

POUG 2026, Krakow, Poland, September 4-5.2026



Rafał Grzegorzczuk
Senior APEX Developer



POUG



ORACLE

Partner



ISO 27001

Oracle SQLcl (SQL Developer Command Line)

- SQLcl is SQL Developer just with CLI vs a GUI
- Over 100 of useful commands
- First, early adopter version was mentioned around 2014/2015
- "SQL*PLus++" , "SDSQL" *
- Modern replacement of SQL Plus
- Latest version 26.2.2 (August 2026)

<https://www.oracle.com/pl/database/sqldeveloper/technologies/sqlcl/>
<https://krisrice.io/2014-12-11-what-is-sdsq/1/>

```
ACCEPT
ALIAS
APEX
APPEND
AQ
ARBORI
ARCHIVE_LOG
ARGUMENT
AWR
BACKGROUND
BLOCKCHAIN_TABLE
BREAK
BRIDGE
BTITLE
CD
CERTIFICATE
CHANGE
CLEAR
CLOUDSTORAGE
CODESCAN
COLUMN
COMPUTE
CONNECT
CONNMG
COPY
CS
CTAS
DATAPUMP
```



About me

- Kielce, Poland - 180 km south from Warsaw
- Oracle APEX and PL/SQL Developer
- Worldwide speaker since 2021
- Oracle ACE Pro

Find me at:

rgrzegorczyk@pretius.com

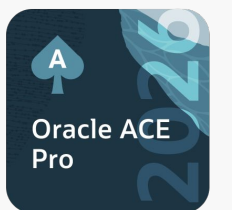
[linkedin.com/in/rgrzegorczyk](https://www.linkedin.com/in/rgrzegorczyk)

<https://rafal.hashnode.dev>



Rafał Grzegorczyk

Senior APEX Developer



Providing what you need, not what you want

pretius

300+

People

60+

APEX Devs

7

Oracle ACEs

70+

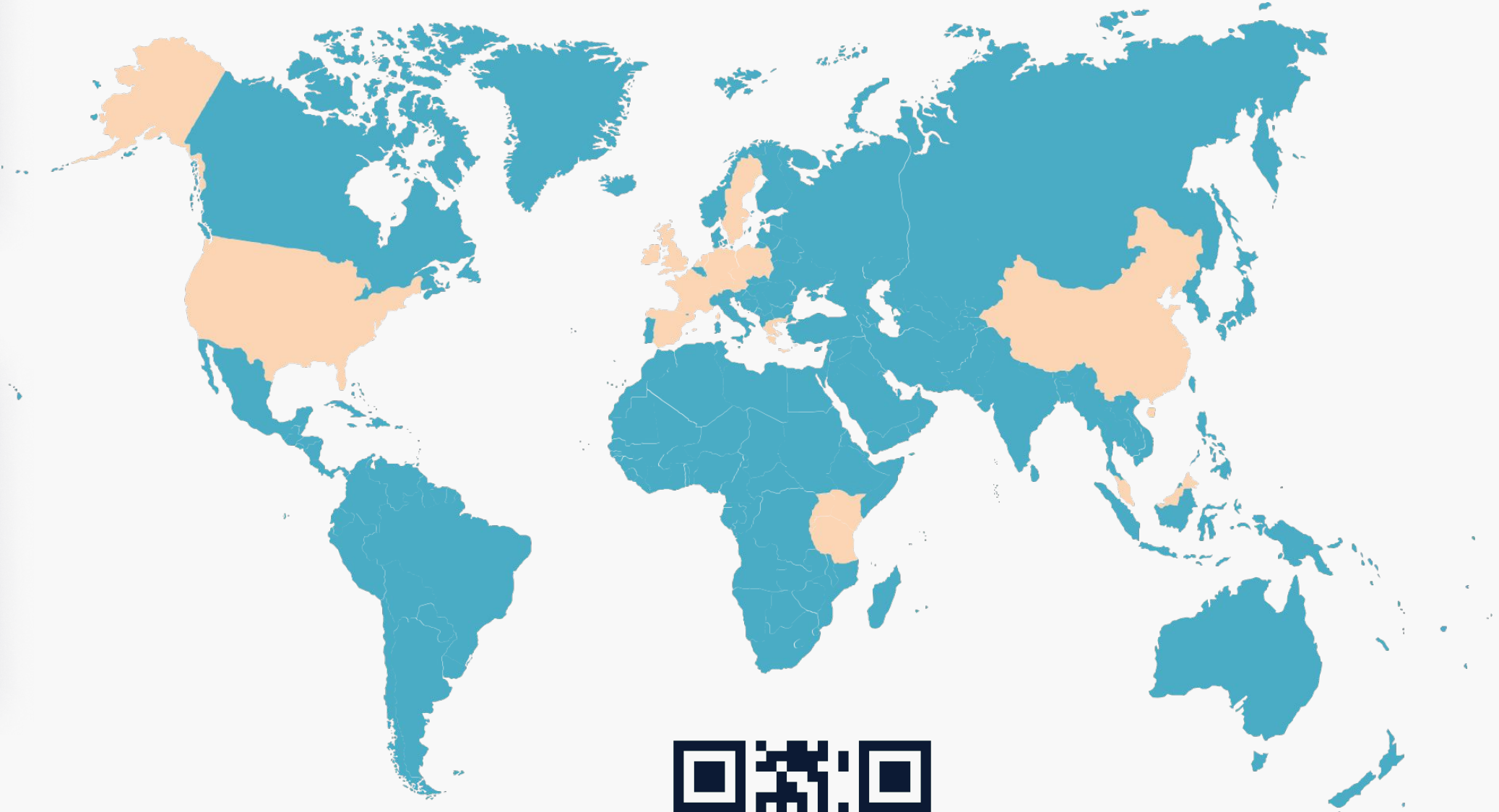
Customers

4

Continents

2

Warsaw, PL
London, UK



Scan to meet us
Pretius.com

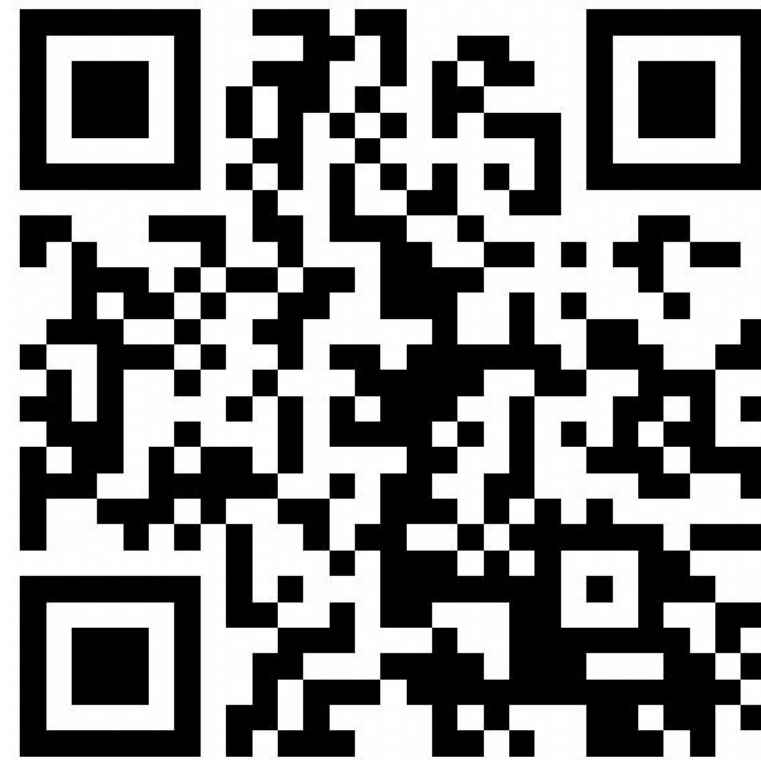


Which tool do you use?



SQLcl features I find most useful:

1. **Model Context Protocol (MCP)** and effortless Skills installation
2. **Aliases** - code snippets / shortcuts
3. **Import/export** data from/to files
4. **Data Pump** - export/import schemas
5. **APEX** and its subcommands (APEXlang included)
6. **CODESCAN** - PL/SQL quality and performance checks
7. **Projects & Liquibase** - complete solution for database & application CI/CD
8. **Schema DIFF** - compare and remove database schema drifts



**GitHub - rgrzegorzcyk/conferences: A place
for public content related to my
presentations at various conferences**

github.com

<https://github.com/rgrzegorzcyk/conferences>



Audience Q&A



SQLcl Installation

pretius

Questions? Ask here:



Do you have SQL Developer Extension in Visual Studio Code?

EXTENSIONS: INSTALLED

@installed SQL Developer



Oracle SQL Developer ...

End-to-end development tools fo...

Oracle Corporation

117ms

Extension: Oracle SQL Developer Extension for VSCode



Oracle SQL Developer Extension for VSCode

Oracle Corporation  oracle.com | 758,601 |  (115)

End-to-end development tools for your SQL and PL/SQL applications.

Disable

Uninstall

☒ Auto Update

DETAILS

FEATURES

CHANGELOG

Oracle SQL Developer Extension for VS Code

About the Oracle SQL Developer extension for VS Code

The gold standard for Oracle Database development and administration is now seamlessly integrated into Visual Studio Code

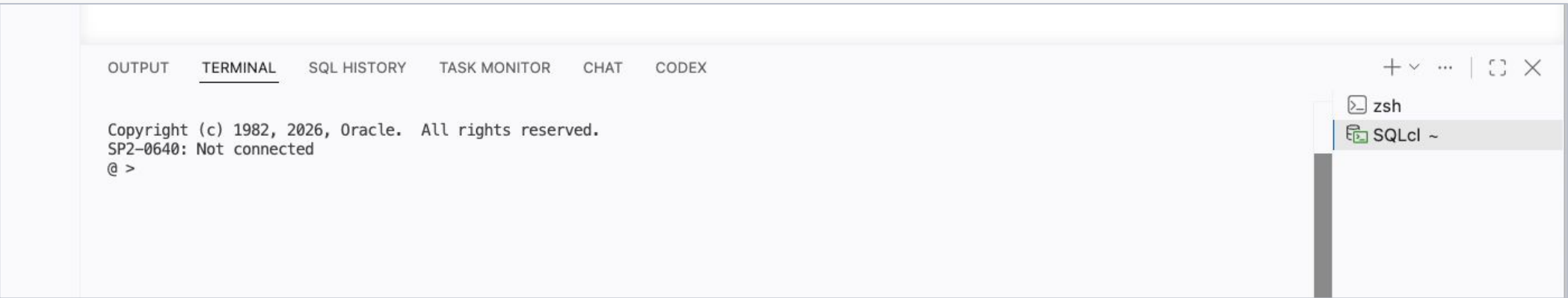
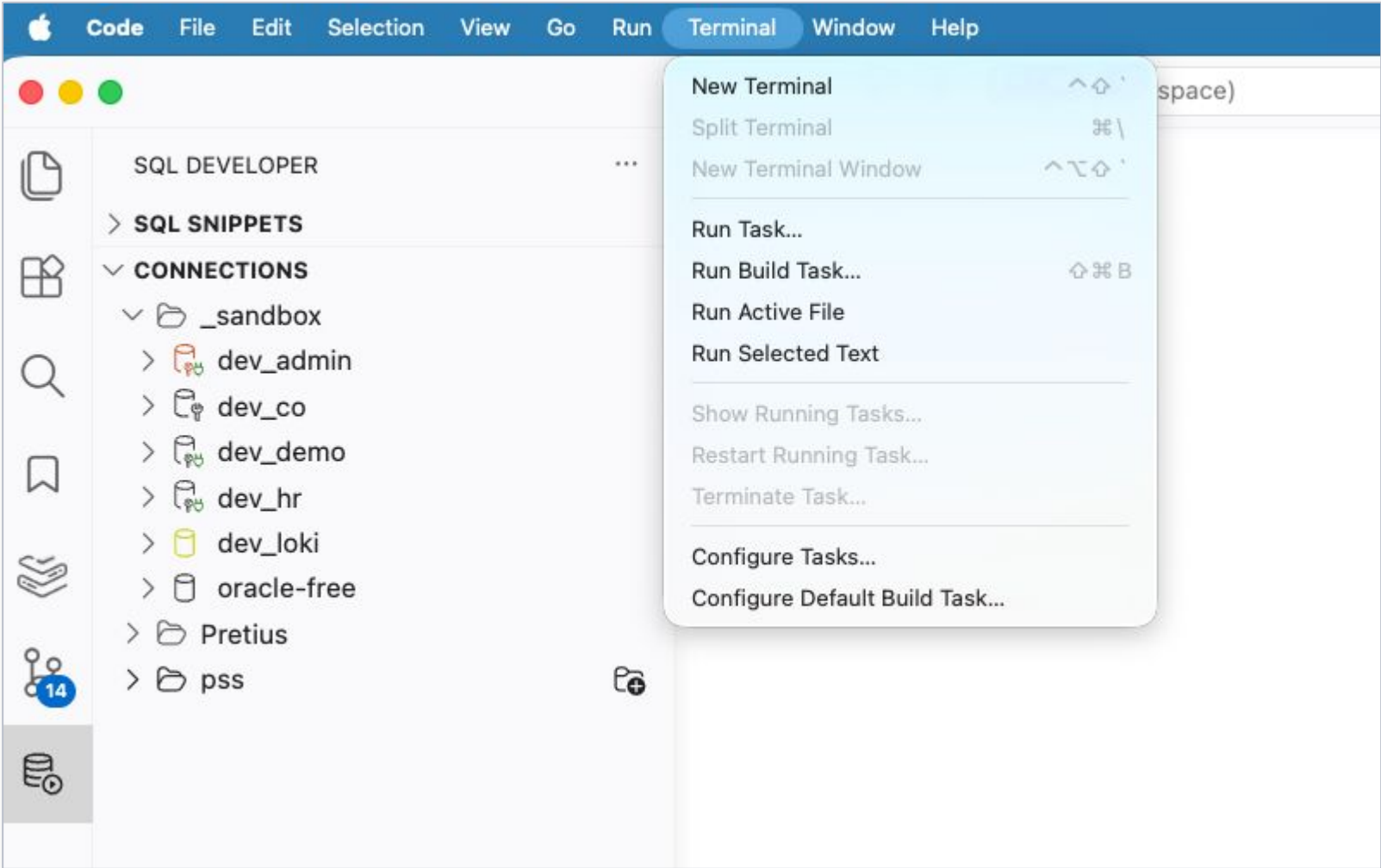
Installation

Identifier	oracle.sql-developer
Version	25.4.1
Last Updated	2 weeks ago
Size	493.92MB

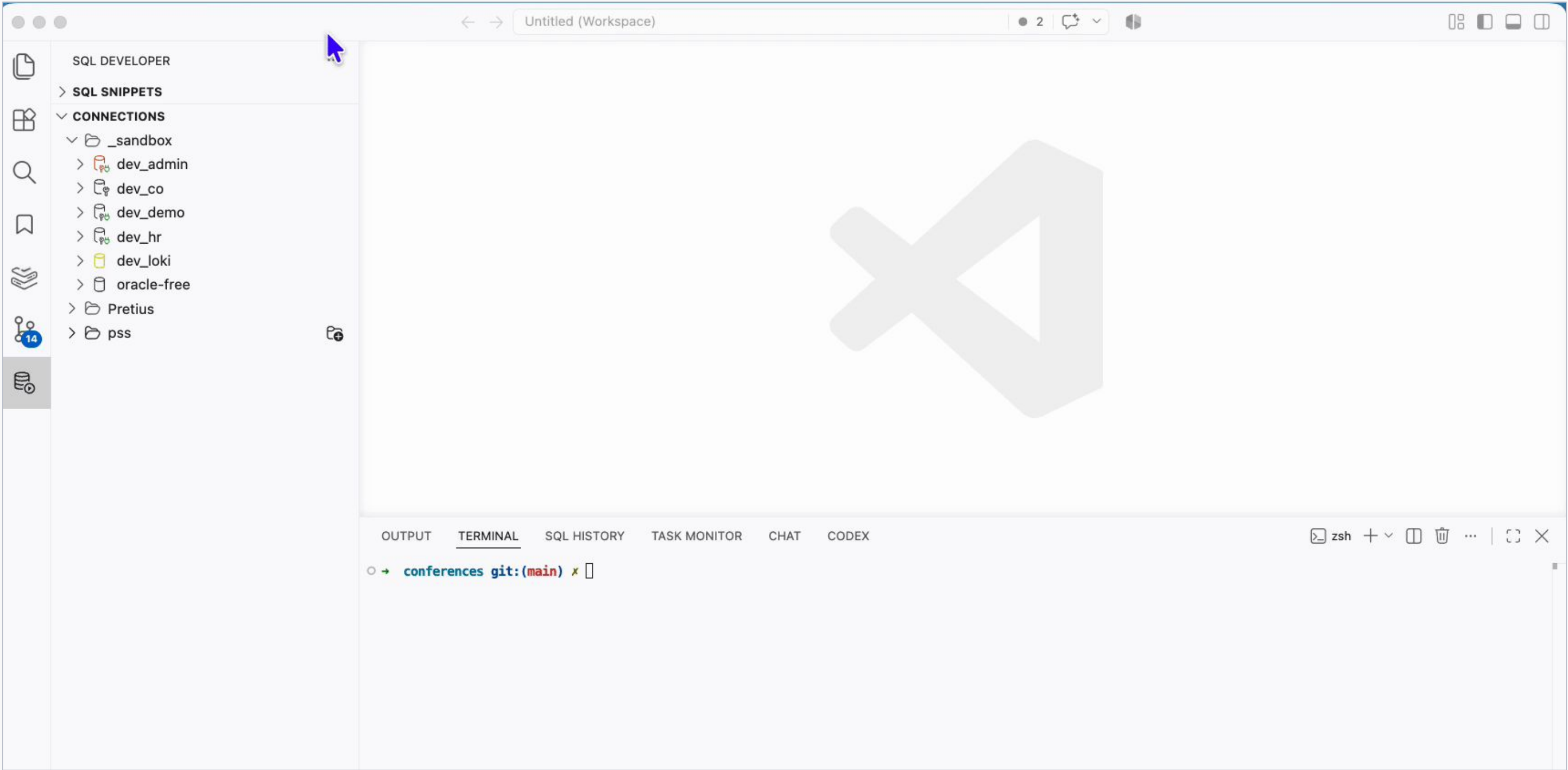


Installation...

SQLcl is already there



SQLcl is already there

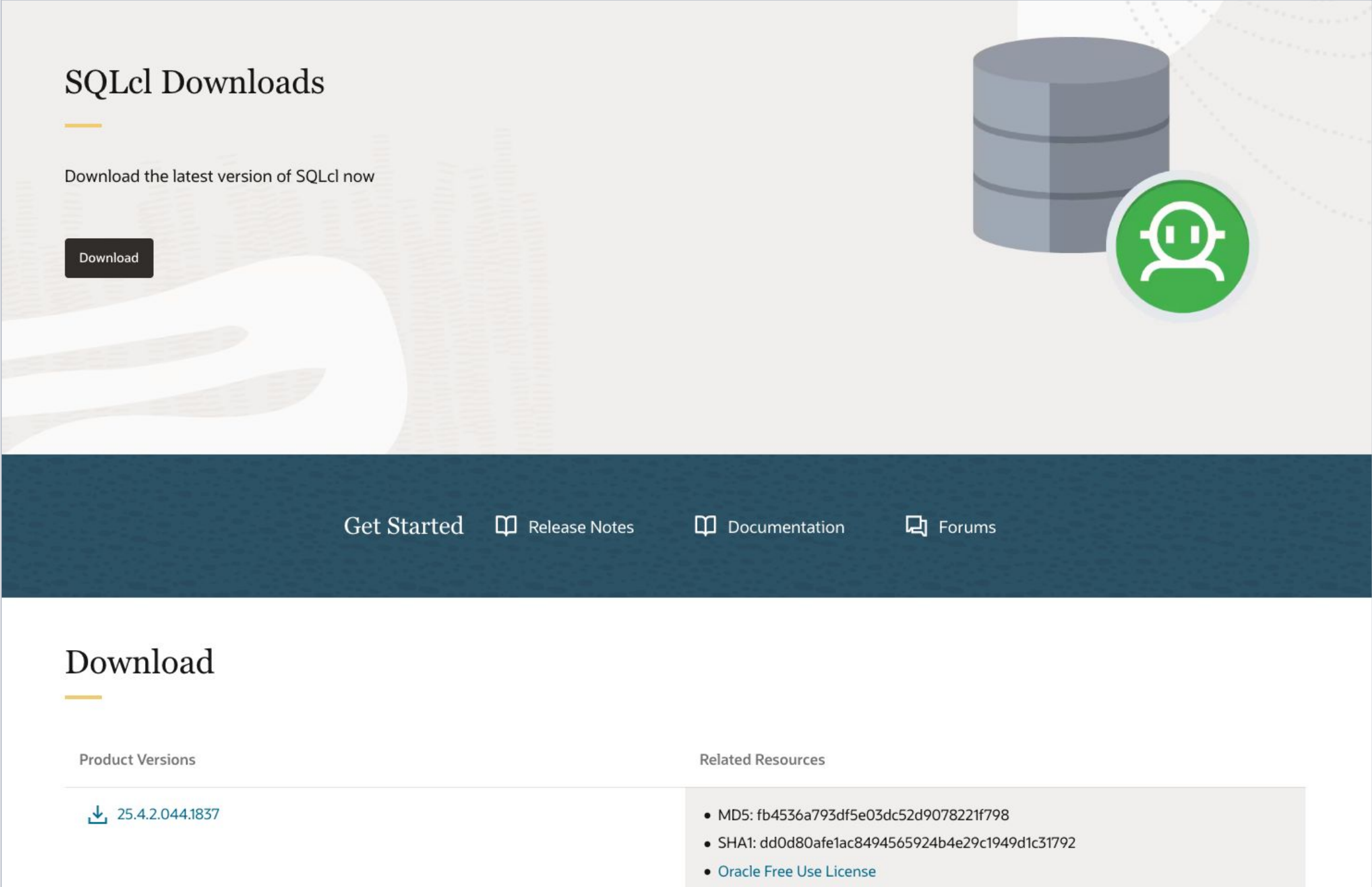


Installation...

SQLcl downloads

For standalone version

go to **www.oracle.com/sqlcl**



The screenshot shows the Oracle SQLcl Downloads page. At the top, there's a header with the title "SQLcl Downloads" and a sub-header "Download the latest version of SQLcl now". A prominent "Download" button is visible. To the right, there's a graphic of a database cylinder and a green circular icon with a robot face. Below the header, a dark blue navigation bar contains links for "Get Started", "Release Notes", "Documentation", and "Forums". The main content area is titled "Download" and is divided into two sections: "Product Versions" and "Related Resources". Under "Product Versions", there's a download link for version 25.4.2.044.1837. Under "Related Resources", there are links for MD5, SHA1, and the Oracle Free Use License.

SQLcl Downloads

Download the latest version of SQLcl now

[Download](#)

[Get Started](#) [Release Notes](#) [Documentation](#) [Forums](#)

Download

Product Versions

[25.4.2.044.1837](#)

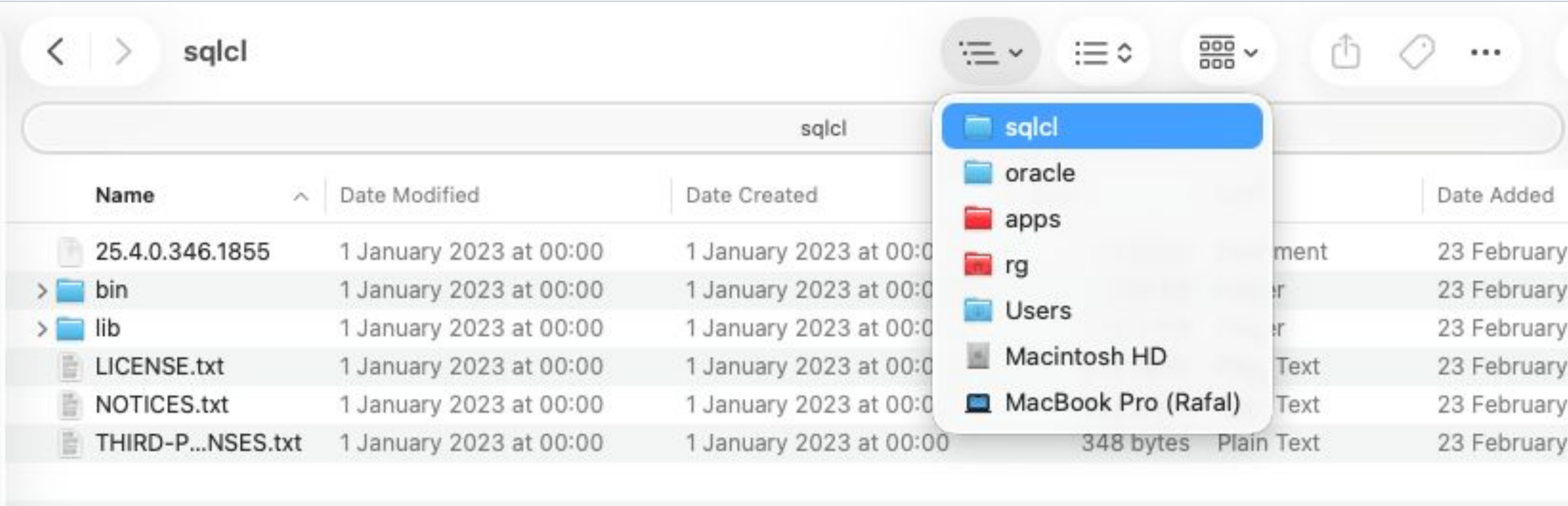
Related Resources

- MD5: fb4536a793df5e03dc52d9078221f798
- SHA1: dd0d80afe1ac8494565924b4e29c1949d1c31792
- [Oracle Free Use License](#)



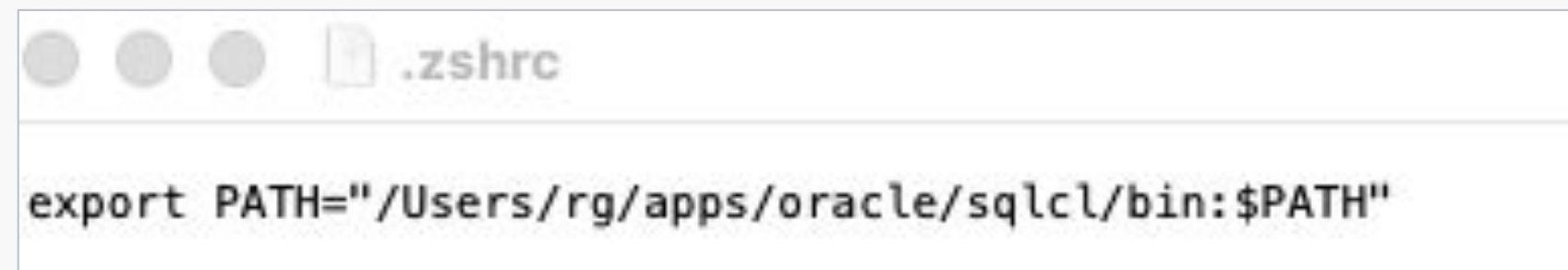
Installation...

Unzip downloaded file and add to chosen folder



Add SQLcl "bin" folder to system path (MacOS)

Users/<your_user>/ .zshrc (or .bash_profile)

A screenshot of a text editor window titled ".zshrc". The window contains the following text:

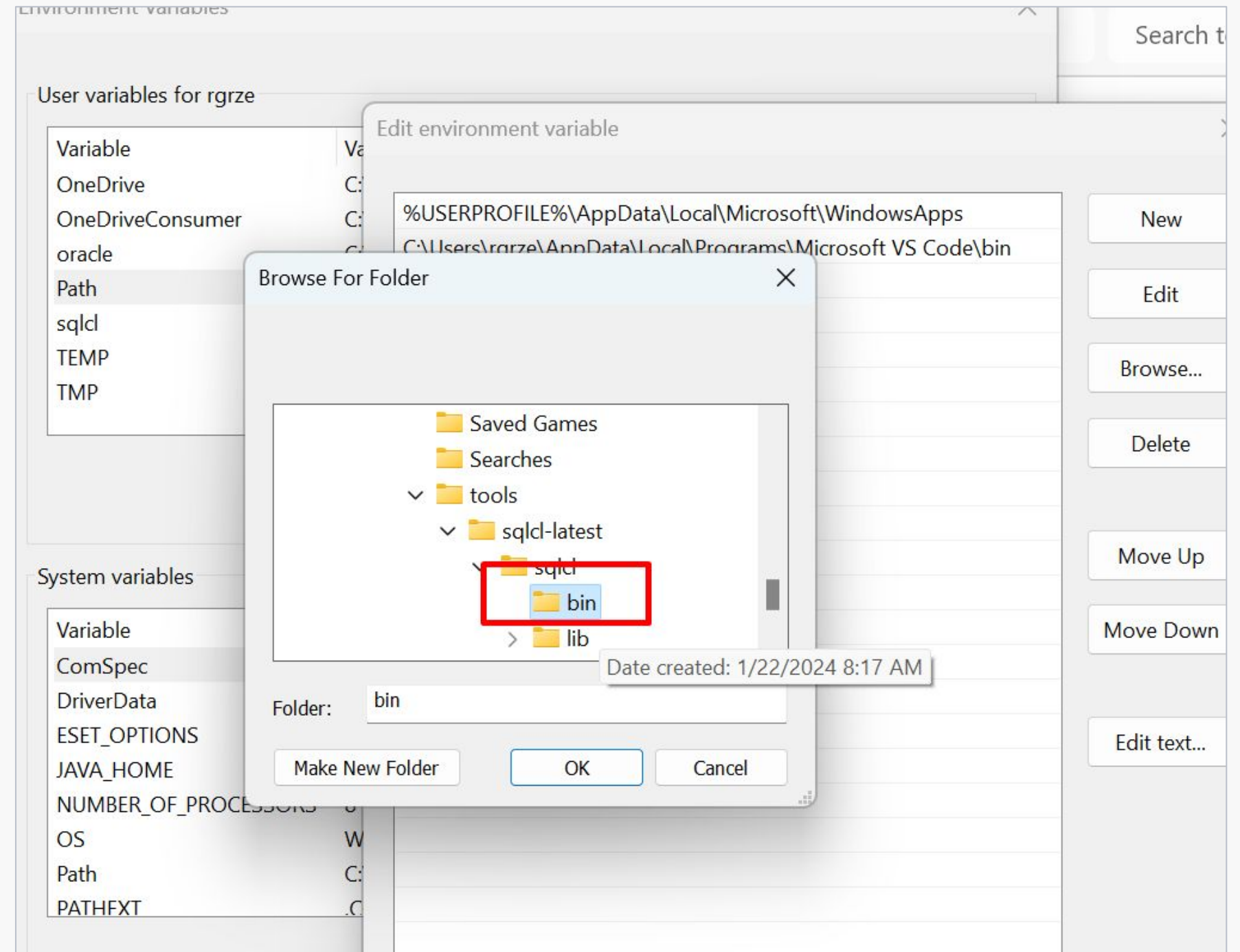
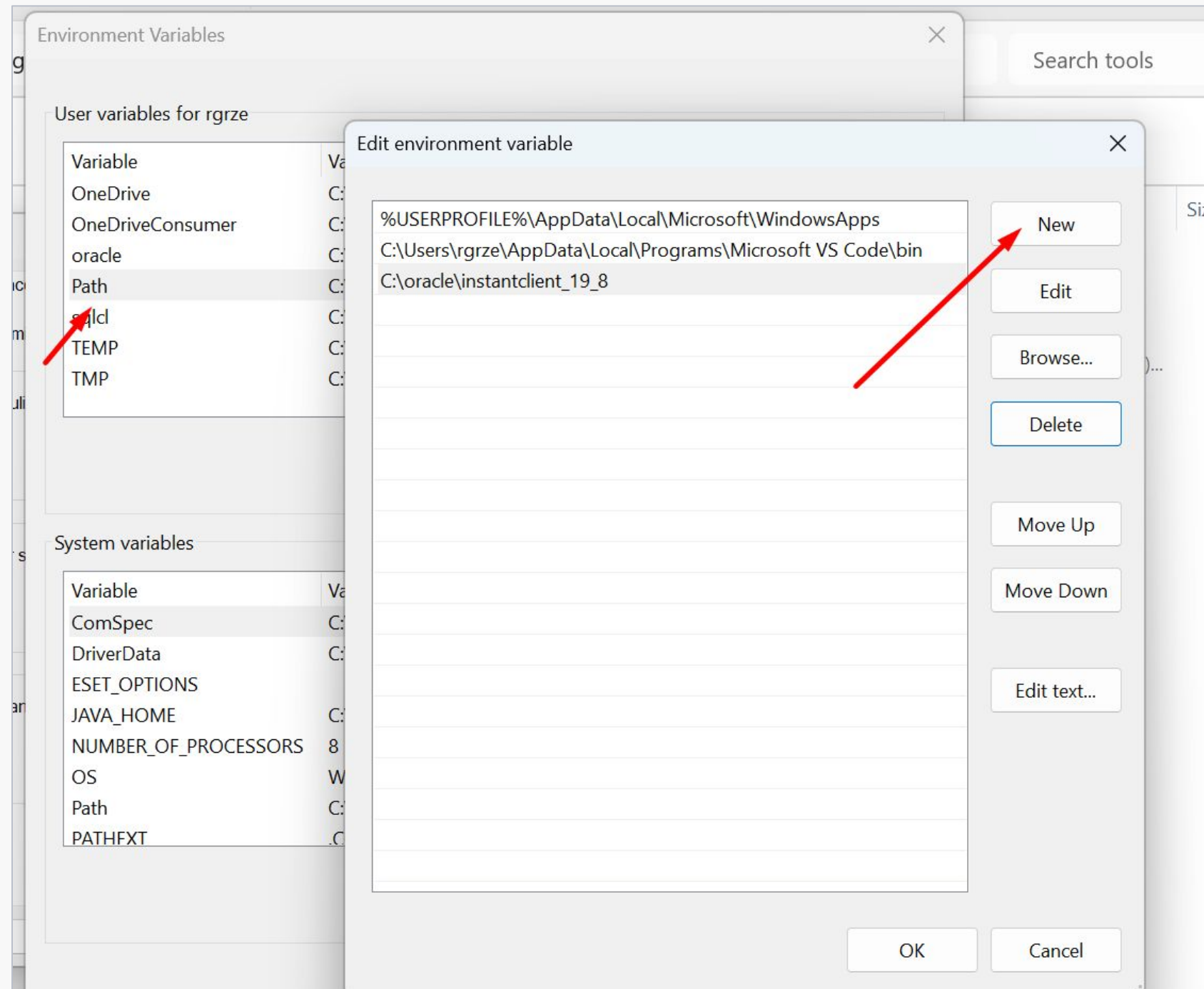
```
export PATH="/Users/rg/apps/oracle/sqlcl/bin:$PATH"
```

```
export PATH="/Users/rg/apps/oracle/sqlcl/bin:$PATH"
```

```
export PATH="/Users/rg/apps/oracle/sqlcl/bin:$PATH"
```



Add SQLcl "bin" folder to system path (Windows)



Verify installation (JAVA is required)

> java -version

TERMINAL

```
● → DEMO java -version
java version "25.0.2" 2026-01-20 LTS
Java(TM) SE Runtime Environment (build 25.0.2+10-LTS-69)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.2+10-LTS-69, mixed mode, sharing)
○ → DEMO
```

> sql -version

TERMINAL

```
● → DEMO sql -version
SQLcl: Release 25.4.0.0 Production Build: 25.4.0.346.1855
○ → DEMO
```

More about
installation [here:](#)



Installation...

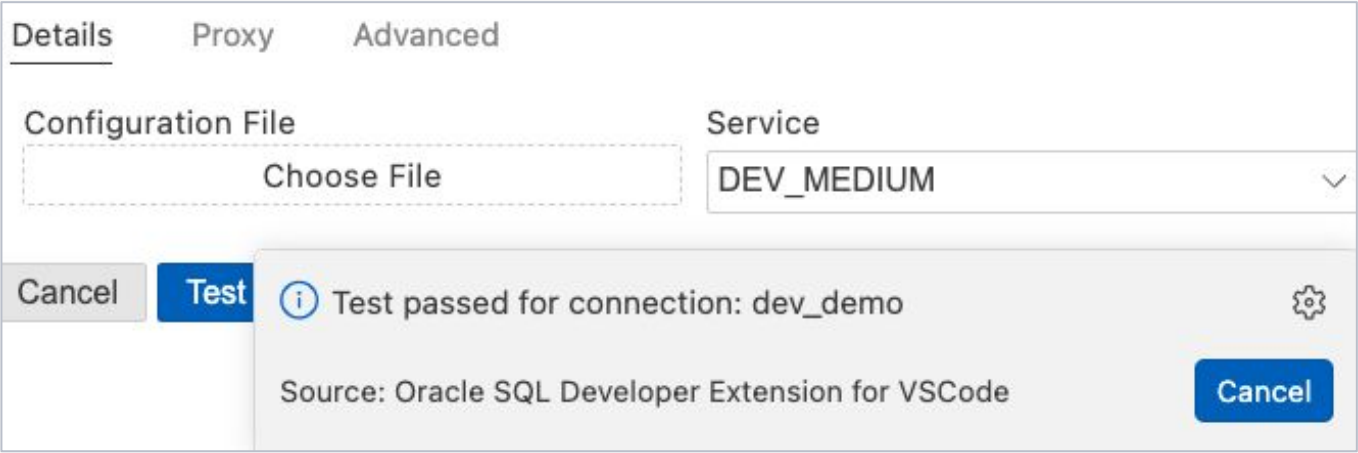
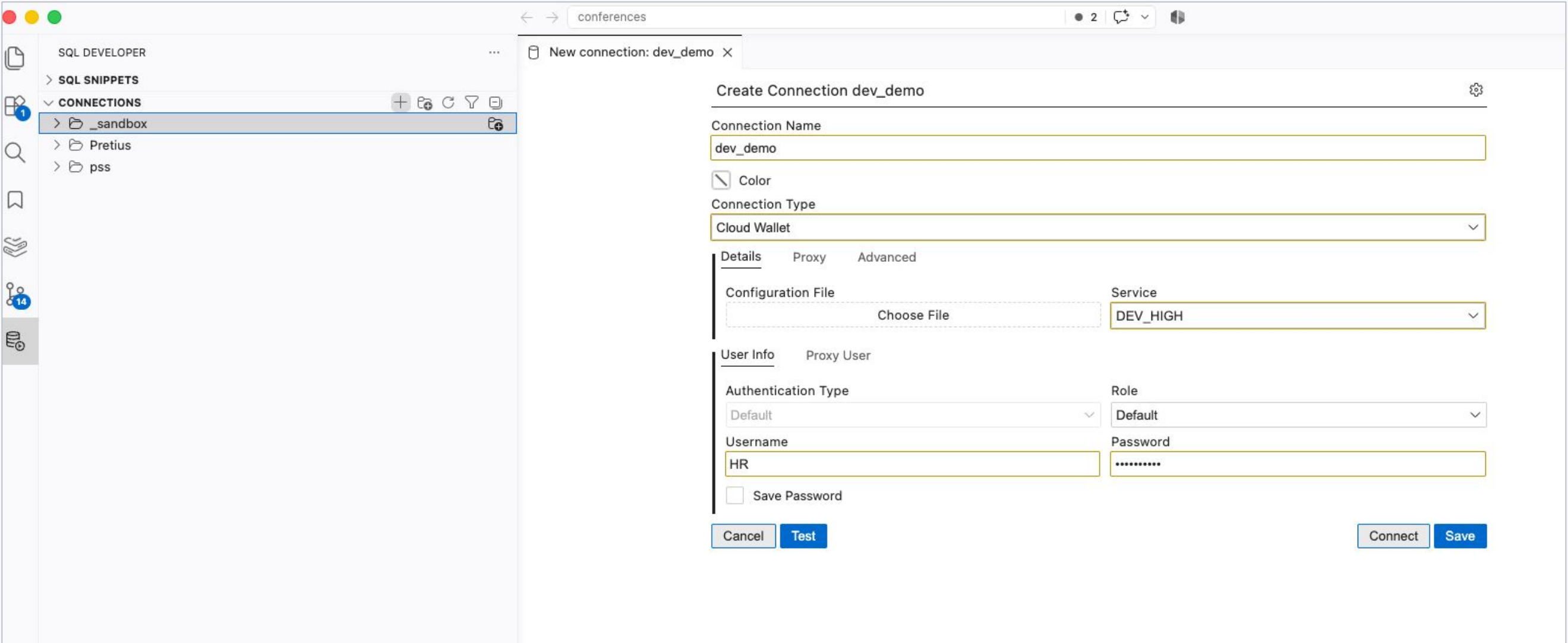
SQLcl connect to database

pretius

Questions? Ask here:

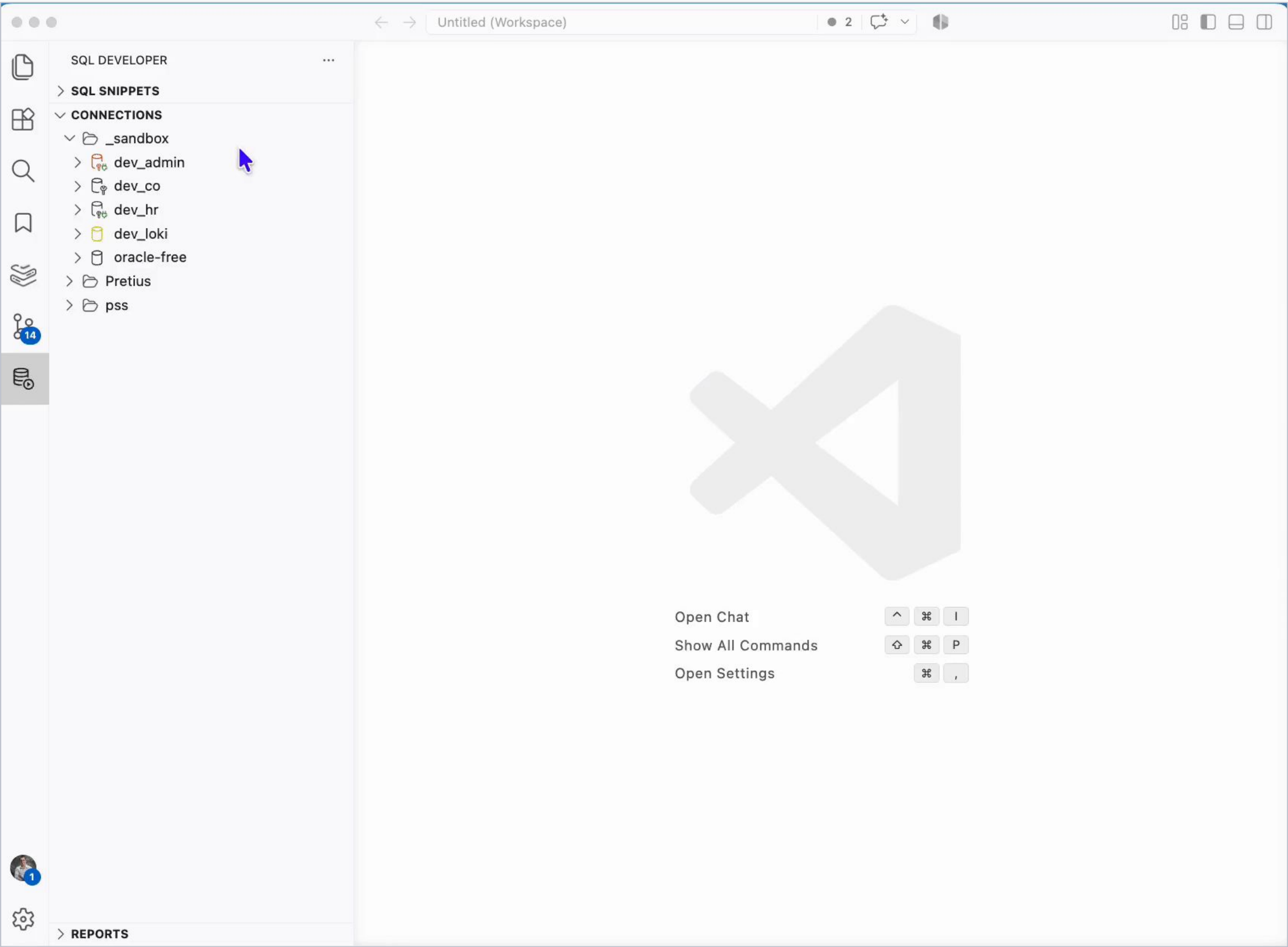


Visual Studio Code SQL Developer Extension (easiest way)



Add connection

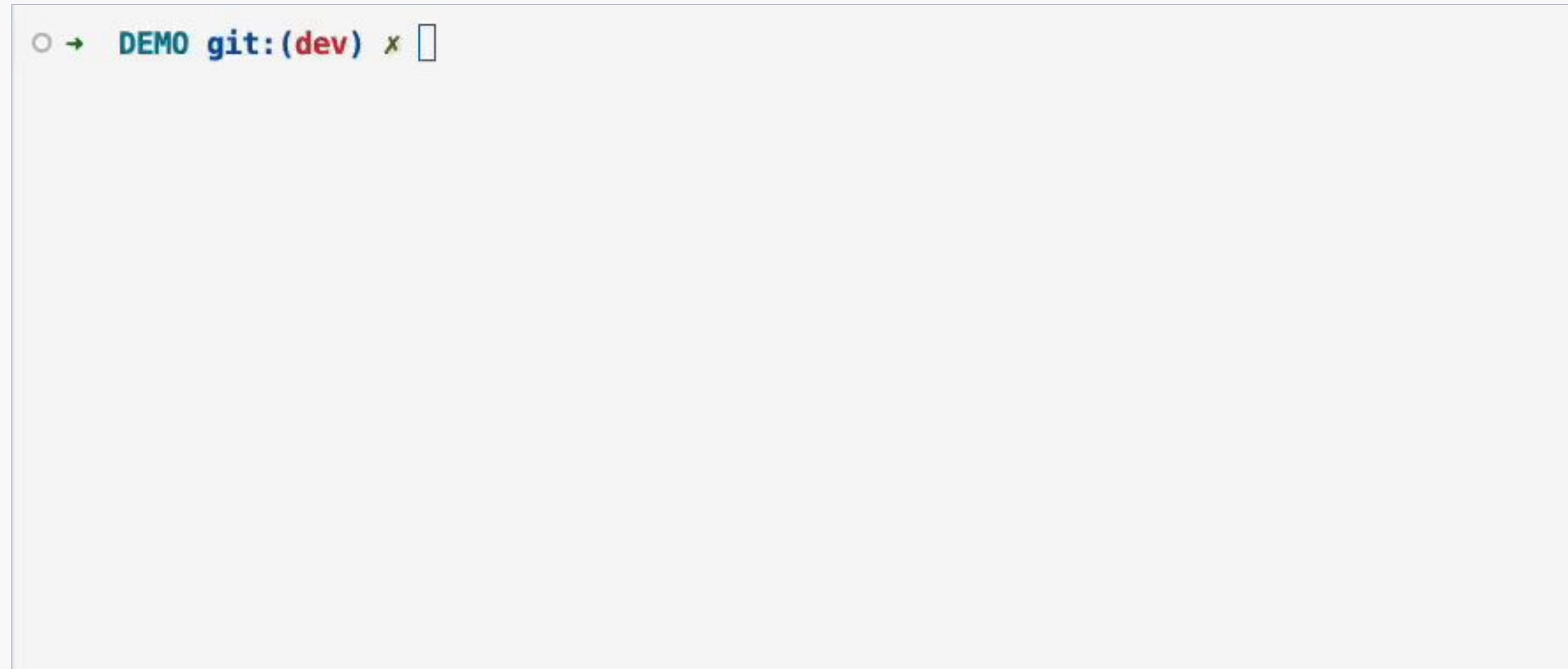
Visual Studio Code SQL Developer Extension (easiest way)



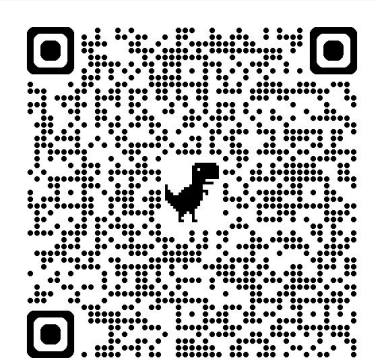
Add connection

Using SQLcl commands (also easy)

> conn -save



More about
connections [here:](#)

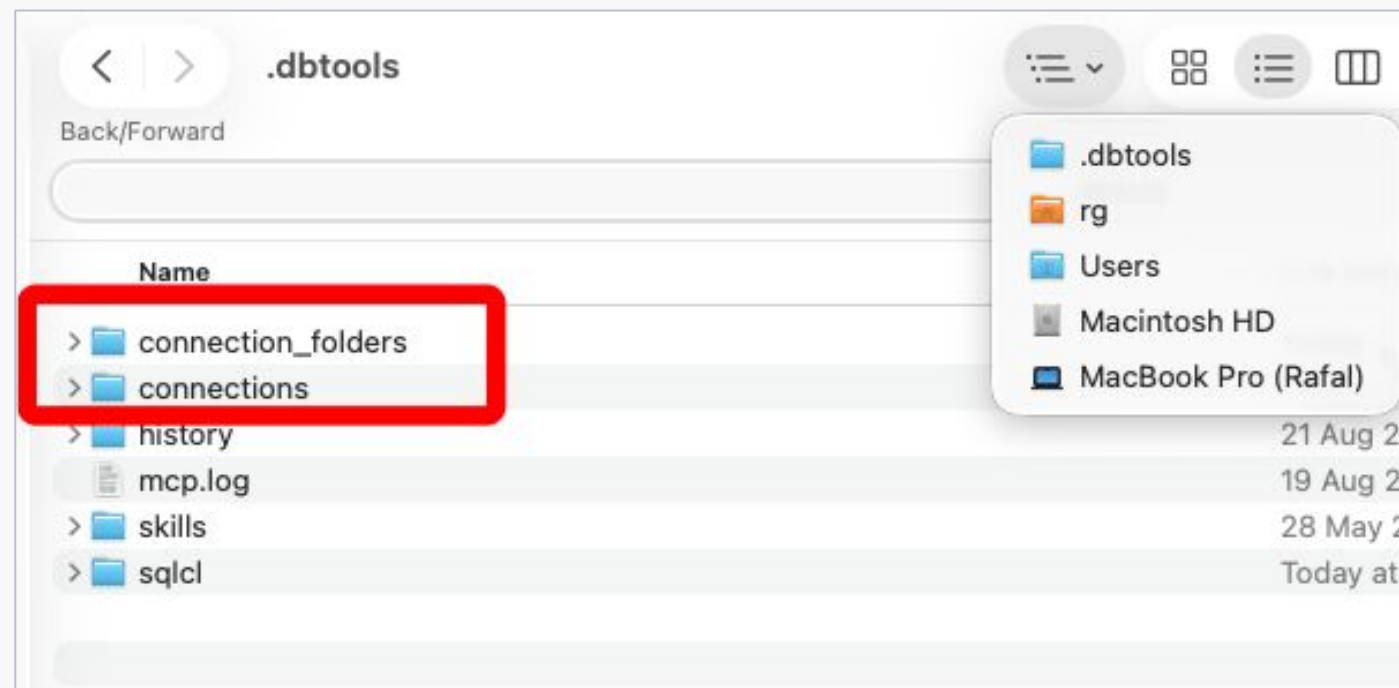


Add connection

Where are my saved connections?

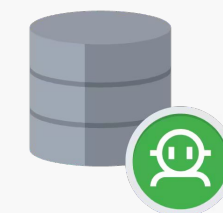
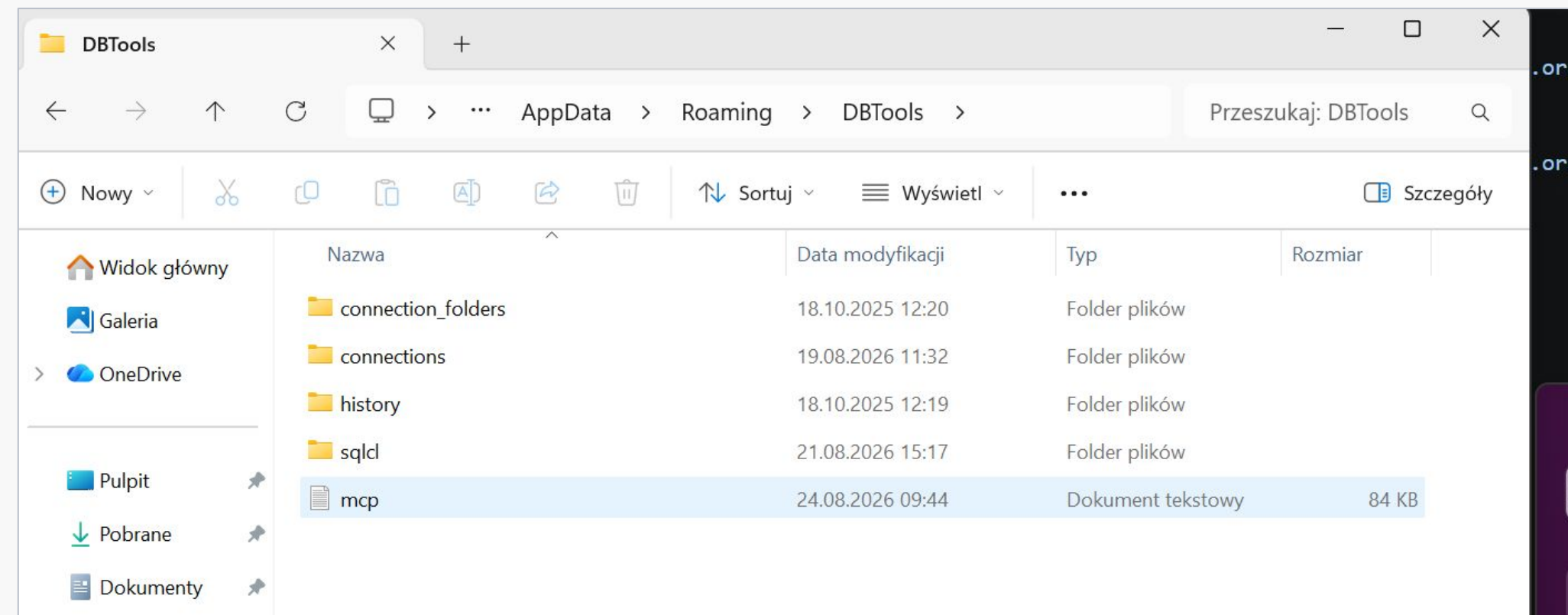
MacOS

/Users/<your_username>/.dbtools



Windows

C:\Users\<username>\AppData\Roaming\DBTools



Add connection

Import/export existing connections (with passwords!)

```
> secret set mySecret MySuperSecret2137!  
> connmgr export -key mySecret -folder /_sandbox sandbox  
  
> secret set mySecret MySuperSecret2137!  
> connmgr import -key mySecret sandbox
```

You can export connections from standalone SQL Developer too !

More about importing/exporting connections:

- Tomas Kucharzyk's blog -> [here](#)
- Jeff Smith's blog -> [here](#)



Add connection

Connect to DB

```
> sql -name <connection_name>
```

TERMINAL

○ → **DEMO** sql -name dev_demo

SQLcl: Release 25.4 Production on Tue Mar 10 10:41:56 2026

Copyright (c) 1982, 2026, Oracle. All rights reserved.

Connected to:

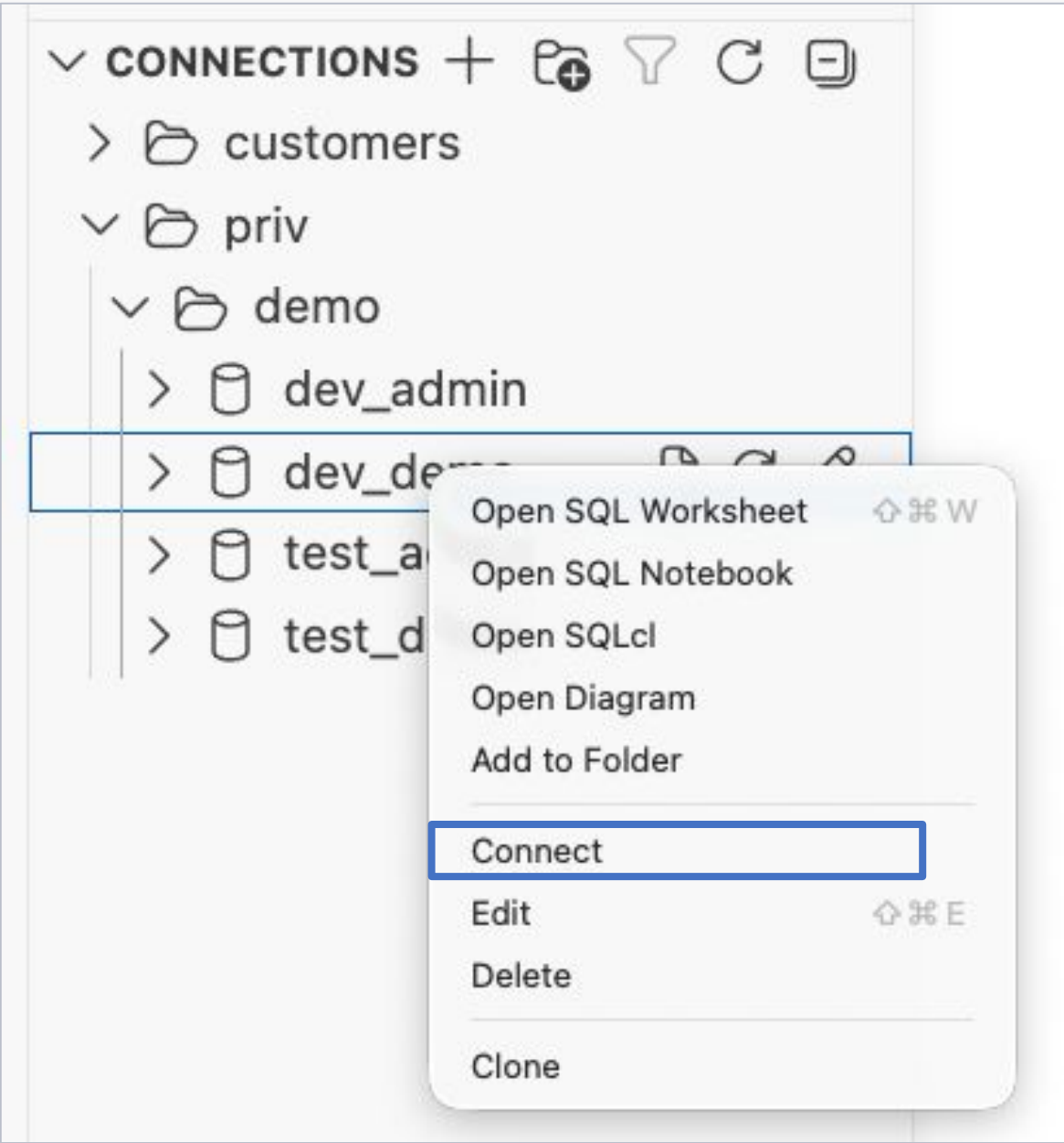
Oracle AI Database 26ai Enterprise Edition Release 23.26.1.2.0 – Production
Version 23.26.1.2.0

SQL>

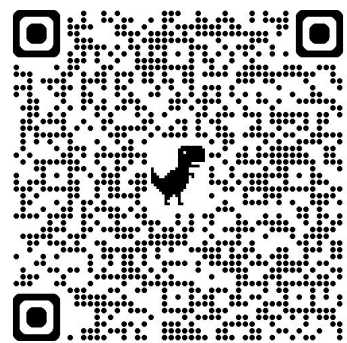


Connect to DB

Using VS Code SQL Developer extension



More about connections [here:](#)



Connect to DB

Before we continue... help...

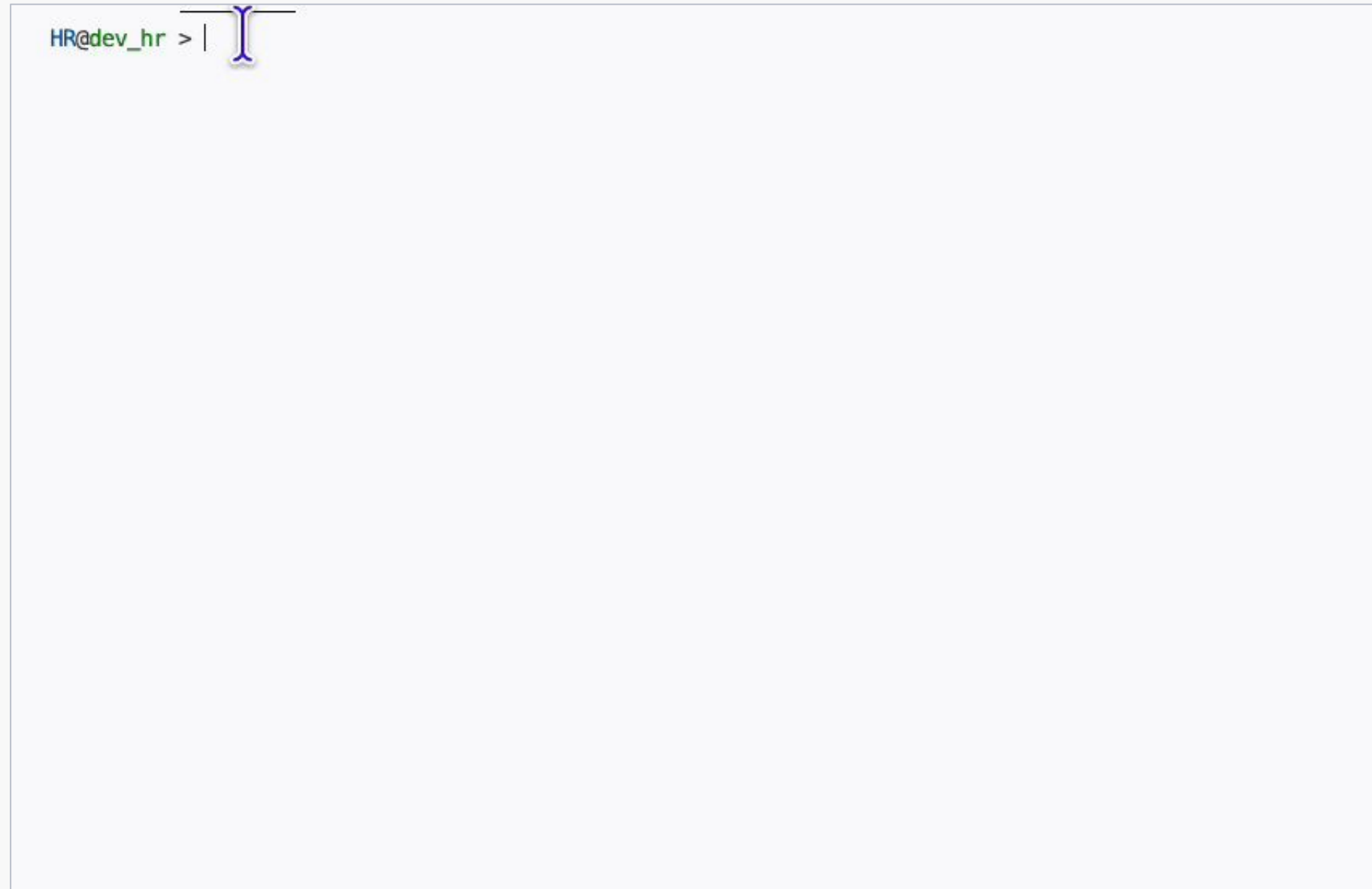
pretius

Questions? Ask here:

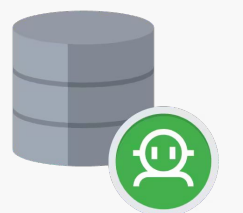


“HELP” is your best friend in SQLcl

```
> help [command] [examples]
```



```
* type "q" to QUIT help
```



HELP

But if you want it AI way...

Ask your AI to explain a command using SQLcl built-in help

OUTPUTTERMINALSQL HISTORYTASK MONITORCHATCODEX

SQLCL PROJECT COMMAND EXPLANATION

Use SQLcl help and explain to me how project command works?

I'll pull the Oracle database skill guidance first, then run SQLcl help for the project command so I can explain it accurately from the official command output.

Read skill db

Next I'll locate SQLcl command docs in the skill files to cross-check terminology, then I'll run SQLcl help so the explanation matches exactly what SQLcl reports.

Searched for files matching project in sqlcl directory

I'll query SQLcl directly with its built-in help for project so I can give you a precise, command-level explanation instead of a generic summary.

Ran Run SQLcl Command - SQLcl - SQL Developer (MCP Server)

I have the SQLcl help output and I'm reading it now to extract the key commands, flow, and practical examples for you.

Reviewed 3 files

SQLcl project is the built-in CI/CD workflow command for database and APEX changes. From help project, it is designed to move work through this lifecycle:

- project init
Creates a new project structure and config (name, schemas, directory, connection name, optional package metadata).
- project export
Exports DB objects (including specific APEX app IDs like APEX.100) into source form in your repo.
- project stage
Builds Liquibase changelogs/changesets from changed SQL files and branch diffs.
- project release
Promotes current staged work into a versioned release (-version required), with optional include/exclude change IDs.
- project gen-artifact
Packages a release into a deployable artifact (zip or tgz).
- project deploy
Deploys an artifact to a target DB (runs install flow from artifact, including Liquibase installer process).



ALIAS - code snippets (shortcuts)

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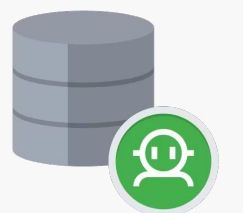
Questions? Ask here:



> alias

Allows to save sql or pl/sql scripts as reusable shortcut commands

Because you can't remember everything :)



Alias

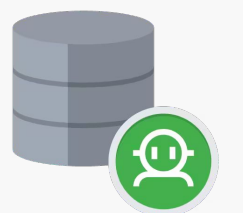
Instead of querying for your SYSDATE each time

```
HR@dev_hr > select sysdate from dual;  
  
SYSDATE  
-----  
25-AUG-26  
  
HR@dev_hr > 
```

Define alias "time" > **alias** time = select sysdate from dual;

```
HR@dev_hr > alias time = select sysdate from dual;  
Alias time added
```

```
HR@dev_hr > time  
  
SYSDATE  
-----  
25-AUG-26  
  
HR@dev_hr > 
```

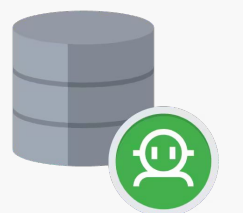


Alias

Instead of querying your top X employees each time

```
HR@dev_hr > select last_name, first_name, salary
              from employees
              where rownum <= 3
              order by salary desc;
```

LAST_NAME	FIRST_NAME	SALARY
King	Steven	24000
Garcia	Lex	17000
Yang	Neena	17000



Define alias "top"

And just type "top" or "top X"

```

HR@dev_hr > top

```

LAST_NAME	FIRST_NAME	SALARY
King	Steven	24000
Yang	Neena	17000
Garcia	Lex	17000
Singh	John	14000
Partners	Karen	13500
Martinez	Michael	13000
Gruenberg	Nancy	12008
Higgins	Shelley	12008
Errazuriz	Alberto	12000
Ozer	Lisa	11500

```

10 rows selected.

HR@dev_hr > top 7

```

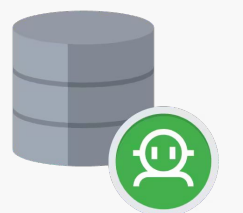
LAST_NAME	FIRST_NAME	SALARY
King	Steven	24000
Yang	Neena	17000
Garcia	Lex	17000
Singh	John	14000
Partners	Karen	13500
Martinez	Michael	13000
Higgins	Shelley	12008

```

7 rows selected.

HR@dev_hr > |

```



Alias

How to define alias?

```
HR@dev_hr > alias -nulldefaults -group employees -desc "My top X employees" top =  
select * from (select last_name, first_name, salary from employees order by salary  
desc) where rownum <= nvl(:count, 10);  
Alias top added
```

> alias

-nulldefaults

- optional, auto-assign null values

-group

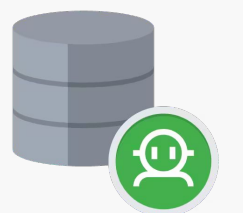
- optional, adds alias to group

-desc

- optional, adds alias description

-<alias_name> = query

- required



Alias

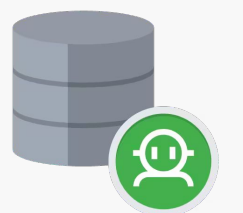
Useful alias subcommands

- > **alias list** • shows user defined aliases

```
HR@dev_hr > alias list
time
top employees - My top X employees
```

- > **alias list all** • shows all aliases

```
HR@dev_hr > alias list all
top - My top employees sorted by salary
top_employees
functions default - functions - show functions from schema
indexes default - indexes - show indexes from schema
locks default - locks <USERNAME>
mview_logs default - mview_logs - show materialized view logs from schema
mviews default - mviews - show materialized views from schema
oraversion default - oraversion - Current Database Version
packages default - packages - show packages from schema
procedures default - procedures - show procedures from schema
```



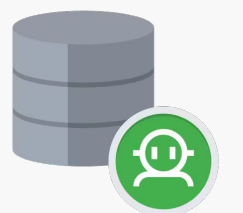
Useful alias subcommands

- > **alias list** <group>
 - shows user defined aliases from grup

```
HR@dev_hr > alias list employees
top employees – My top X employees
```

- > **alias details** <alias_name>
 - shows alias details

```
HR@dev_hr > alias details top
top NULLDEFAULTS employees – My top X employees
-----
count: count
-----
select * from (select last_name, first_name, salary from employees order by salary desc) where rownum <= nvl(:count, 10)
```



Useful alias subcommands

> **alias** search *text*

- * are optional wildcards

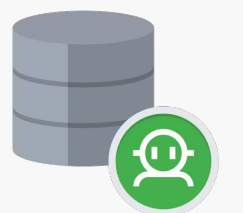
```
HR@dev_hr > alias search salary
top
HR@dev_hr > alias search *last*
RepDBAJobs      RepWaitClassActivityforLastHour  RepYourJobs      top
HR@dev_hr >
```

> **alias** save

- save aliases

> **alias** load

- load aliases shared by friend



Alias

Useful aliases you already have!

> **alias list default**

```

HR@dev_hr > alias list default
functions      default      - functions - show functions from schema
indexes        default      - indexes - show indexes from schema
locks          default      - locks <USERNAME>
mview_logs     default      - mview_logs - show materialized view logs from schema
mviews         default      - mviews - show materialized views from schema
oraversion     default      - oraversion - Current Database Version
packages       default      - packages - show packages from schema
procedures     default      - procedures - show procedures from schema
queues         default      - queues - show queues from schema
schedules      default      - schedules - show schedules from schema
sequences      default      - sequences - show sequences from schema
sessions       default
synonyms        default      - synonyms - show synonyms from schema
tables          default      - tables - show tables from schema
triggers        default      - triggers - show triggers from schema
types           default      - types - show types from schema
u_jobs          default      - jobs - show jobs from schema
u_mle_modules   default      - mle_modules - show mle_modules from schema
views           default      - views - show views from schema

```

```

HR@dev_hr > 

```



Export/Import data using SQLcl

pretius

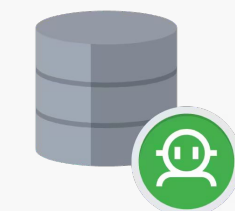
Questions? Ask here:



> set sqlformat insert

```
SQL> set sqlformat insert
SQL> select name,country,rating from eba_demo_ig_people fetch first 10 rows only;
REM INSERTING into EBA_DEMO_IG_PEOPLE
SET DEFINE OFF;
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Keizo Obuchi','Japan',22.475646);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Antonín Novotný','Czech Republic',22.474758);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Shivaji','India',22.474212);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Frank Rijkaard','Netherlands',22.473873);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Richard Pryor','United States',22.473754);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Nico','Germany',22.473584);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Alexander II of Scotland','United Kingdom',22.473236);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Jacques Anquetil','France',22.472601);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Hideki Shirakawa','Japan',22.47257);
Insert into EBA_DEMO_IG_PEOPLE (NAME,COUNTRY,RATING) values ('Chet Atkins','United States',22.472027);

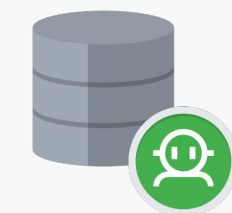
10 rows selected.
```



Export data

```
> set sqlformat json
```

```
SQL> set sqlformat json
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
{"results":[{"columns":[{"name":"NAME","type":"VARCHAR2"}, {"name":"COUNTRY","type":"VARCHAR2"}, {"name":"RATING","type":"NUMBER"}], "items":
[
{"name":"Giuseppe Farina","country":"Italy","rating":22.153414}
,{"name":"Geena Davis","country":"United States","rating":22.152783}
,{"name":"Bengt I. Samuelsson","country":"Sweden","rating":22.152342}
,{"name":"Susan Hayward","country":"United States","rating":22.151325}
,{"name":"Grand Duchess Maria Nikolaevna of Russia","country":"Russia","rating":22.151002}
,{"name":"Chang Myon","country":"South Korea","rating":22.149127}
,{"name":"Vladimir Voronin","country":"Moldova","rating":22.148992}
,{"name":"P. W. Botha","country":"South Africa","rating":22.148361}
,{"name":"Hans Sloane","country":"United Kingdom","rating":22.146437}
,{"name":"Howard Aiken","country":"United States","rating":22.14543}
]
]
}}
10 rows selected.
```

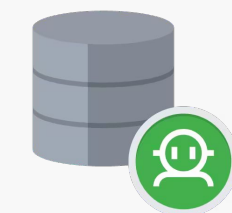


Export data

```
> set sqlformat csv
> spool [file_name]
> spool off
```

```
SQL> set feedback off
SQL> set sqlformat csv
SQL> spool my_csv_file.csv
SQL> select name, country, rating from eba_demo_ig_people fetch first 10 rows only;
"NAME","COUNTRY","RATING"
"Keizo Obuchi","Japan",22.475646
"Antonín Novotný","Czech Republic",22.474758
"Shivaji","India",22.474212
"Frank Rijkaard","Netherlands",22.473873
"Richard Pryor","United States",22.473754
"Nico","Germany",22.473584
"Alexander II of Scotland","United Kingdom",22.473236
"Jacques Anquetil","France",22.472601
"Hideki Shirakawa","Japan",22.47257
"Chet Atkins","United States",22.472027
SQL> spool off;
```

my_csv_file.csv X	
my_csv_file.csv > data	
1	"NAME","COUNTRY","RATING"
2	"Johann Sebastian Bach","Germany",30.167197
3	"Ludwig van Beethoven","Germany",30.109331
4	"Cleopatra VII of Egypt","Egypt",30.036677
5	"Galileo Galilei","Italy",29.96003
6	"Euclid","Unknown",29.884558
7	"Nero","Italy",29.859917
8	"Karl Marx","Germany",29.842897
9	"Marco Polo","Italy",29.771886
10	"Cicero","Italy",29.742594
11	"Vincent van Gogh","Netherlands",29.738438
12	



Export data

- > set loadformat csv
- > unload sample_restaurant

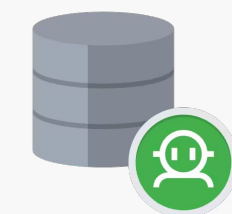
```
SQL> set loadformat csv
SQL> unload sample_restaurant

format csv

column_names on
delimiter ,
enclosures ""
encoding UTF8
row_terminator default

** UNLOAD Start ** at 2026.03.17-08.46.50
Export Separate Files to /Users/rg/DEMO
DATA TABLE SAMPLE_RESTAURANT
File Name: /Users/rg/DEMO/SAMPLE_RESTAURANT_DATA_TABLE.csv
Number of Rows Exported: 11
** UNLOAD End ** at 2026.03.17-08.46.51
SQL> |
```

SAMPLE_RESTAURANT_DATA_TABLE.csv	
SAMPLE_RESTAURANT_DATA_TABLE.csv > data	
1	"ID","NAME","ADDRESS","RATING","LOGO_URL","HERO_URL","CATEGORY_ID","CREATED"
2	1,"ByteStart Breakfast Nook","Address #123","4.0","restaurant/bytestart-br
3	2,"SQLicious Bistro","Address #724","3.2","restaurant/sqlicious-bistro.jpg"
4	3,"PL/SQLoely Diner","Address #335","5.0","restaurant/sqloely-diner.jpg"
5	4,"Oracool Cafe","Address #294","4.5","restaurant/oracool-cafe.jpg","resta
6	5,"APEXquisite Delights","Address #513","4.3","restaurant/apexquisite-deli
7	6,"SQLtastic Grill","Address #367","3.7","restaurant/sqltastic-grill.jpg","
8	7,"PL/SQL Paradise","Address #211","4.9","restaurant/sql-paradise.jpg","re
9	8,"APEXtraordinary Fare","Address #841","2.9","restaurant/apextraordinary-
10	9,"Oraculicious Cuisine","Address #751","4.7","restaurant/oraculicious-cuis
11	10,"APEX Mex Grill","Address #571","5.0","restaurant/apex-mex-grill.jpg","
12	11,"SQL Sushi Bar","Address #382","4.7","restaurant/sql-sushi-grill.jpg","
13	



Export data

- > set loadformat json
- > unload sample_restaurant

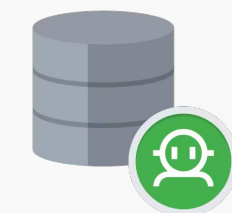
```
SQL> set loadformat json
SQL> unload sample_restaurant

format json

** UNLOAD Start ** at 2026.03.17-08.51.04
Export Separate Files to /Users/rg/DEMO
DATA TABLE SAMPLE_RESTAURANT
File Name: /Users/rg/DEMO/SAMPLE_RESTAURANT_DATA_TABLE.json
Number of Rows Exported: 11
** UNLOAD End ** at 2026.03.17-08.51.04
SQL> 
```

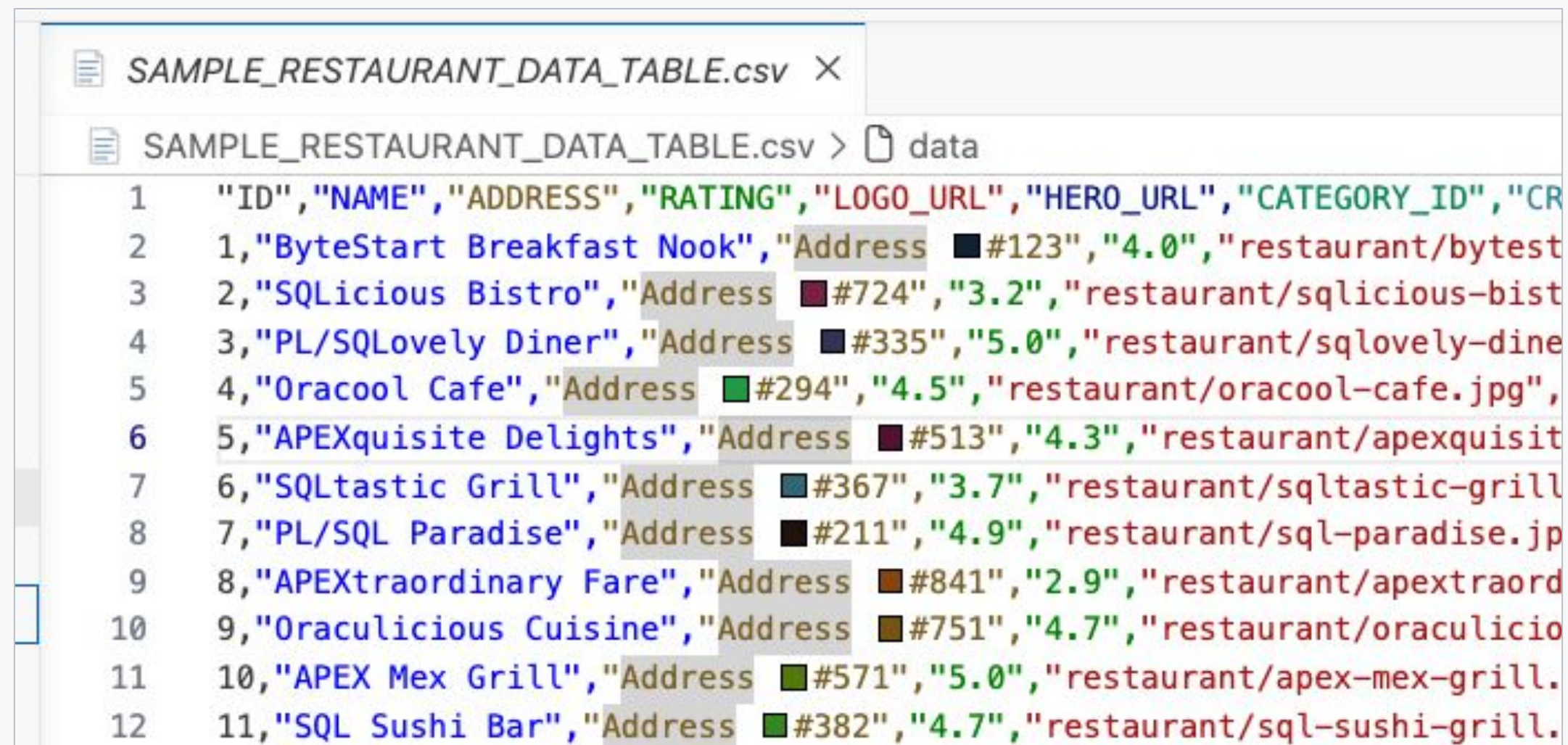
```
{ } SAMPLE_RESTAURANT_DATA_TABLE.json X
{ } SAMPLE_RESTAURANT_DATA_TABLE.json > ...

1  {"results":[{"columns":[{"name":"ID","type":"NUMBER"}, {"name":"NAME"}
2  [
3  {"id":1,"name":"ByteStart Breakfast Nook","address":"Address #123",
4  , {"id":2,"name":"SQLicious Bistro","address":"Address #724","rating"
5  , {"id":3,"name":"PL\SQLovely Diner","address":"Address #335","ratin
6  , {"id":4,"name":"Oracool Cafe","address":"Address #294","rating":"4.
7  , {"id":5,"name":"APEXquisite Delights","address":"Address #513","rat
8  , {"id":6,"name":"SQLtastic Grill","address":"Address #367","rating":
9  , {"id":7,"name":"PL\SQL Paradise","address":"Address #211","rating"
10 , {"id":8,"name":"APEXtraordinary Fare","address":"Address #841","rat
11 , {"id":9,"name":"Oraculicious Cuisine","address":"Address #751","rat
12 , {"id":10,"name":"APEX Mex Grill","address":"Address #571","rating":
13 , {"id":11,"name":"SQL Sushi Bar","address":"Address #382","rating":
14 ]}]}
```

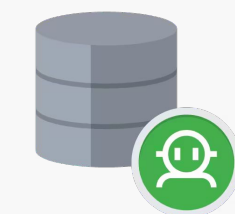


Export data

- > **load** [table_name] [file_name] [options]
- > **load** conference sample_restaurant_data_table.csv **NEW**



1	"ID"	"NAME"	"ADDRESS"	"RATING"	"LOGO_URL"	"HERO_URL"	"CATEGORY_ID"	"CR"
2	1	"ByteStart Breakfast Nook"	"Address #123"	"4.0"	"restaurant/bytestart-breakfast-nook.jpg"	"restaurant/bytestart-breakfast-nook.jpg"	"1"	"CR"
3	2	"SQLicious Bistro"	"Address #724"	"3.2"	"restaurant/sqlicious-bistro.jpg"	"restaurant/sqlicious-bistro.jpg"	"2"	"CR"
4	3	"PL/SQL Lovely Diner"	"Address #335"	"5.0"	"restaurant/sql-lovely-diner.jpg"	"restaurant/sql-lovely-diner.jpg"	"3"	"CR"
5	4	"Oracool Cafe"	"Address #294"	"4.5"	"restaurant/oracool-cafe.jpg"	"restaurant/oracool-cafe.jpg"	"4"	"CR"
6	5	"APEXquisite Delights"	"Address #513"	"4.3"	"restaurant/apexquisite-delights.jpg"	"restaurant/apexquisite-delights.jpg"	"5"	"CR"
7	6	"SQLtastic Grill"	"Address #367"	"3.7"	"restaurant/sqltastic-grill.jpg"	"restaurant/sqltastic-grill.jpg"	"6"	"CR"
8	7	"PL/SQL Paradise"	"Address #211"	"4.9"	"restaurant/sql-paradise.jpg"	"restaurant/sql-paradise.jpg"	"7"	"CR"
9	8	"APEXtraordinary Fare"	"Address #841"	"2.9"	"restaurant/apextraordinary-fare.jpg"	"restaurant/apextraordinary-fare.jpg"	"8"	"CR"
10	9	"Oraculicious Cuisine"	"Address #751"	"4.7"	"restaurant/oraculicious-cuisine.jpg"	"restaurant/oraculicious-cuisine.jpg"	"9"	"CR"
11	10	"APEX Mex Grill"	"Address #571"	"5.0"	"restaurant/apex-mex-grill.jpg"	"restaurant/apex-mex-grill.jpg"	"10"	"CR"
12	11	"SQL Sushi Bar"	"Address #382"	"4.7"	"restaurant/sql-sushi-grill.jpg"	"restaurant/sql-sushi-grill.jpg"	"11"	"CR"



Import data

```
SQL> load conference sample_restaurant_data_table.csv NEW
```

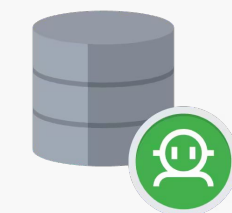
```
csv
column_names on
delimiter ,
enclosures ""
double off
encoding UTF8
row_limit off
row_terminator default
skip_rows 0
skip_after_names
```

```
Create new table and load data into table DEMO.CONFERENCE
```

```
#INFO DATE format detected: DD-MON-RR
```

```
CREATE TABLE DEMO.CONFERENCE
(
  ID NUMBER(4),
  NAME VARCHAR2(128),
  ADDRESS VARCHAR2(26),
  RATING NUMBER(4, 1),
  LOGO_URL VARCHAR2(128),
  HERO_URL VARCHAR2(128),
  CATEGORY_ID NUMBER(3),
  CREATED DATE,
  CREATED_BY VARCHAR2(26),
  UPDATED DATE,
  UPDATED_BY VARCHAR2(26),
  DELIVERY_FEE NUMBER(3),
  DELIVERY_TIME NUMBER(4)
)
;

#INFO Table created
#INFO Number of rows processed: 11
#INFO Number of rows in error: 0
#INFO Last row processed in final committed batch: 11
SUCCESS: Processed without errors
SQL> |
```



Import data

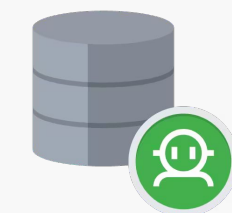
> **load** conference sample_restaurant_data_table.csv

```
SQL> load conference sample_restaurant_data_table.csv
```

```
csv  
column_names on  
delimiter ,  
enclosures ""  
double off  
encoding UTF8  
row_limit off  
row_terminator default  
skip_rows 0  
skip_after_names
```

```
Load data into table DEMO.CONFERENCE
```

```
#INFO Number of rows processed: 11  
#INFO Number of rows in error: 0  
#INFO Last row processed in final committed batch: 11  
SUCCESS: Processed without errors  
SQL> |
```



Import data

SQLcl Data Pump

pretius

Questions? Ask here:



Export full schema from DEV [schema DEMO] [on-premise]

> dp export

```
SQL> dp export
Initiating DATA PUMP
Database Time Zone: VERSION:44 CON_ID:0
Log Location: DATA_PUMP_DTR:ESQL_770.LOG
Starting "DEMO"."ESQL_770":
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA/LOGREP
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_BODY
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCOBJ/SCHEDULER
Processing object type SCHEMA_EXPORT/SEQUENCE/SEQUENCE
Processing object type SCHEMA_EXPORT/TABLE/COMMENT
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/IDENTITY_COLUMN
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/PROCEDURE
Processing object type SCHEMA_EXPORT/PACKAGE/COMPILE_PACKAGE/PACKAGE_SPEC/ALTER_PACKAGE_SPEC
Processing object type SCHEMA_EXPORT/FUNCTION/ALTER_FUNCTION
Processing object type SCHEMA_EXPORT/PROCEDURE/ALTER_PROCEDURE
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX
Processing object type SCHEMA_EXPORT/VIEW/VIEW
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT
Processing object type SCHEMA_EXPORT/TABLE/TRIGGER
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT
. . exported "DEMO"."ESQL_500"          2.4 MB      2218 rows
. . exported "DEMO"."ESQL_498"          2.2 MB      2196 rows
. . exported "DEMO"."ESQL_496"          2.1 MB      2172 rows
```

```
. . exported "DEMO"."EBA_DEMO_DG_PRODUCTS"      9.3 KB      46 rows
. . exported "DEMO"."EBA_DEMO_MD_COMMENTS"      9.2 KB      11 rows
. . exported "DEMO"."EBA_DEMO_DG_EMP"           9 KB       14 rows
. . exported "DEMO"."EBA_DEMO_TREE_EMP"         9 KB       14 rows
. . exported "DEMO"."EBA_UT_CHART_PROJECTS"      8.7 KB      14 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES" 8.5 KB      10 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS" 8.1 KB       8 rows
. . exported "DEMO"."EBA_DEMO_MD_STATUS"        8 KB        3 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"  6.9 KB      20 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDERS"   6.6 KB       5 rows
. . exported "DEMO"."EBA_DEMO_IG_DEPT"          6.5 KB       4 rows
. . exported "DEMO"."EBA_DEMO_DG_DEPT"          6.1 KB       4 rows
. . exported "DEMO"."EBA_DEMO_TREE_DEPT"        6.1 KB       4 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"  0 KB        0 rows
. . exported "DEMO"."EBA_VECTOR_AI_HELPER"       0 KB        0 rows
. . exported "DEMO"."EBA_VECTOR_DATATABLE"       0 KB        0 rows
ORA-39173: Encrypted data has been stored unencrypted in dump file set.
Master table "DEMO"."ESQL_770" successfully loaded/unloaded
*****
Dump file set for DEMO.ESQL_770 is:
/u03/dbfs/4B003B659121328EE063BB59000ACD83/data/dpdump/ESQL_770.DMP
Job "DEMO"."ESQL_770" successfully completed at Tue Mar 17 16:47:19 2026 elapsed 0 00:01:15
DataPump Operation Status 'COMPLETED'
```



Export full schema from DEV [DEMO] [OCI]

```
> datapump EXPORT -schemas DEMO -directory DATA_PUMP_DIR -credential
DEMO_OBJ_STORAGE -dumpuri MY_BUCKET_URL/demo_schema.dmp
```

```
. . exported "DEMO"."EBA_DEMO_MD_MILESTONES"          10.8 KB      12 rows
. . exported "DEMO"."EBA_DEMO_MD_TASK_LINKS"           11 KB         7 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_ORDERS"          10.1 KB     105 rows
. . exported "DEMO"."EBA_DEMO_MD_PROJECTS"           10.1 KB         5 rows
. . exported "DEMO"."EBA_DEMO_DG_PRODUCTS"             9.3 KB        46 rows
. . exported "DEMO"."EBA_DEMO_MD_COMMENTS"            9.2 KB        11 rows
. . exported "DEMO"."EBA_DEMO_DG_EMP"                  9 KB          14 rows
. . exported "DEMO"."EBA_DEMO_TREE_EMP"                9 KB          14 rows
. . exported "DEMO"."EBA_UT_CHART_PROJECTS"            8.7 KB        14 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES"      8.5 KB        10 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS"    8.1 KB         8 rows
. . exported "DEMO"."EBA_DEMO_MD_STATUS"               8 KB           3 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"        6.9 KB        20 rows
. . exported "DEMO"."SAMPLE_RESTAURANT_ORDERS"         6.6 KB         5 rows
. . exported "DEMO"."EBA_DEMO_IG_DEPT"                 6.5 KB         4 rows
. . exported "DEMO"."EBA_DEMO_DG_DEPT"                 6.1 KB         4 rows
. . exported "DEMO"."EBA_DEMO_TREE_DEPT"              6.1 KB         4 rows
. . exported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"        0 KB           0 rows
. . exported "DEMO"."EBA_VECTOR_AI_HELPER"             0 KB           0 rows
. . exported "DEMO"."EBA_VECTOR_DATATABLE"            0 KB           0 rows
ORA-39173: Encrypted data has been stored unencrypted in dump file set.
Master table "DEMO"."ESQL_772" successfully loaded/unloaded
*****
Dump file set for DEMO.ESQL 772 is:
https://objectstorage[redacted]/demo_schema.dmp
Job "DEMO"."ESQL_772" successfully completed at Tue Mar 17 17:05:51 2026 elapsed 0 00:00:57
DataPump Operation Status 'COMPLETED'
```



Data Pump

Import full schema [DEMO] into TEST [on premise]

> sql -name test_demo

> datapump IMPORT -schemas DEMO -directory DATA_PUMP_DIR -credential
DEMO_OBJ_STORAGE -dumpuri MY_BUCKET_URL/o/demo_schema.dmp

```
SQL> DATAPUMP IMPORT -schemas DEMO -directory DATA_PUMP_DIR -credential DEMO_OBJ_STORAGE -dumpuri /bucket-data-pump-demo/o/demo_schema.dmp
Initiating DATA PUMP
Database Time Zone: VERSION:44 CON_ID:0
Log Location: DATA_PUMP_DIR:ISQL_420.LOG
Master table "DEMO"."ISQL_420" successfully loaded/unloaded
Starting "DEMO"."ISQL_420":
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA/LOGREP
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCOBJ/SCHEDULER
ORA-39083: Object type SCHEDULER:PROCOBJ failed to create with error:
ORA-20022: Credential "DEMO"."DEMO_OBJ_STORAGE" already exists
Processing object type SCHEMA_EXPORT/SEQUENCE/SEQUENCE
Processing object type SCHEMA_EXPORT/TABLE/TABLE
Processing object type SCHEMA_EXPORT/TABLE/TABLE_DATA
. . imported "DEMO"."EBA_DEMO_SEARCH_WEB_CREDS"          0 KB          0 rows
. . imported "DEMO"."EBA_VECTOR_DATATABLE"              0 KB          0 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_CUSTOMERS"         6.9 KB         20 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_ORDERS"          10.1 KB        105 rows
. . imported "DEMO"."EBA_DEMO_DG_DEPT"                 6.1 KB          4 rows
. . imported "DEMO"."SAMPLE_RESTAURANT_ITEMS"          12.8 KB         24 rows
. . imported "DEMO"."EBA_UT_DEMO_CARDS"               19.4 KB         18 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_PRODUCTS"         45.1 KB         46 rows
. . imported "DEMO"."EBA_DEMO_MD_TASKS"                19.1 KB         45 rows
. . imported "DEMO"."SAMPLE_RESTAURANT"               12.1 KB         11 rows
. . imported "DEMO"."EBA_DEMO_DG_EMP"                  9 KB           14 rows
. . imported "DEMO"."EBA_DEMO_MD_TASK_TODOS"          11.2 KB          9 rows
. . imported "DEMO"."EBA_DEMO_IG_DEPT"                 6.5 KB          4 rows
. . imported "DEMO"."EBA_DEMO_MD_PROJECTS"            10.1 KB          5 rows
. . imported "DEMO"."EBA_DEMO_DG_CUSTOMERS"            25 KB         392 rows
. . imported "DEMO"."EBA_DEMO_IG_EMP"                 12.6 KB         15 rows
. . imported "DEMO"."EBA_DEMO_TREE_STOCKS"            11.6 KB         11 rows
. . imported "DEMO"."ESQL_770"                        2.6 MB        2240 rows
. . imported "DEMO"."EBA_DEMO_SEARCH_STORES"          11.6 KB         23 rows
. . imported "DEMO"."ESQL_400"                        2.2 MB        2106 rows
```



Data Pump

Import full schema [DEMO] into TEST [on premise]

```
> sql -name test_demo
```

```
> datapump IMPORT -schemas DEMO -directory DATA_PUMP_DIR -credential  
DEMO_OBJ_STORAGE -dumpuri MY_BUCKET_URL/o/demo_schema.dmp
```

```
. . imported "DEMO"."SAMPLE_RESTAURANT_ORDERS"      6.6 KB      5 rows  
. . imported "DEMO"."SAMPLE_RESTAURANT_ORDER_ITEMS" 8.1 KB      8 rows  
. . imported "DEMO"."EBA_DEMO_TREE_DEPT"            6.1 KB      4 rows  
. . imported "DEMO"."CONFERENCE"                   13.8 KB     22 rows  
. . imported "DEMO"."ESQL_500"                      2.4 MB    2218 rows  
. . imported "DEMO"."EBA_DEMO_MD_COMMENTS"          9.2 KB     11 rows  
. . imported "DEMO"."SAMPLE_RESTAURANT_NEW"         12.2 KB     11 rows  
. . imported "DEMO"."EBA_DEMO_DG_PRODUCTS"          9.3 KB     46 rows  
. . imported "DEMO"."EBA_DEMO_MD_MILESTONES"        10.8 KB     12 rows  
. . imported "DEMO"."EBA_VECTOR_MOVEDATA"           10 KB      10 rows  
. . imported "DEMO"."EBA_DEMO_MD_TEAM_MEMBERS"      11 KB      12 rows  
. . imported "DEMO"."SAMPLE_RESTAURANT_CATEGORIES"  8.5 KB     10 rows  
. . imported "DEMO"."EBA_DEMO_TREE_PROJ_FILES"       51 KB       4 rows  
. . imported "DEMO"."EBA_UT_MAP_AIRPORTS"          20.7 KB     45 rows  
Processing object type SCHEMA_EXPORT/TABLE/COMMENT  
Processing object type SCHEMA_EXPORT/TABLE/IDENTITY_COLUMN  
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_SPEC  
Processing object type SCHEMA_EXPORT/FUNCTION/FUNCTION  
Processing object type SCHEMA_EXPORT/PROCEDURE/PROCEDURE  
Processing object type SCHEMA_EXPORT/PACKAGE/COMPILE_PACKAGE/PACKAGE_SPEC/ALTER_PACKAGE_SPEC  
Processing object type SCHEMA_EXPORT/FUNCTION/ALTER_FUNCTION  
Processing object type SCHEMA_EXPORT/PROCEDURE/ALTER_PROCEDURE  
Processing object type SCHEMA_EXPORT/VIEW/VIEW  
Processing object type SCHEMA_EXPORT/PACKAGE/PACKAGE_BODY  
Processing object type SCHEMA_EXPORT/TABLE/INDEX/INDEX  
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/CONSTRAINT  
Processing object type SCHEMA_EXPORT/TABLE/INDEX/STATISTICS/INDEX_STATISTICS  
Processing object type SCHEMA_EXPORT/TABLE/CONSTRAINT/REF_CONSTRAINT  
Processing object type SCHEMA_EXPORT/TABLE/TRIGGER  
Processing object type SCHEMA_EXPORT/TABLE/STATISTICS/TABLE_STATISTICS  
Processing object type SCHEMA_EXPORT/POST_SCHEMA/PROCACT_SCHEMA/LBAC_EXP  
Job "DEMO"."ISQL_420" completed with 1 error(s) at Tue Mar 17 17:25:42 2026 elapsed 0 00:00:55  
DataPump Operation Status 'COMPLETED'
```



Data Pump

SHOW LEVEL & SCRIPTS

(new commands since May 2026)

Questions? Ask here:

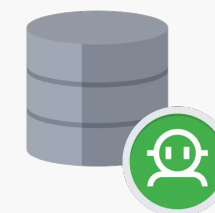


> show level

Displays the current script nesting depth and current script path

> show scripts

Displays the current script call stack



Show level & scripts

EXPLORER

> OPEN EDITORS

DEMO_SQLCL

> apex

> show_level_and_scripts

> includes

> level1

> level2

> level3

> level4

> level5

level5.sql

level4.sql

level3.sql

level2.sql

level1.sql

demo.md

main.sql

README.md

main.sql U X

show_level_and_scripts > main.sql > ...

1 prompt === main.sql start ===

2 show scripts

3 show level

4 select 'my MAIN level query' as script_name from dual;

5 @@includes/level1/level1.sql

level1.sql U X

show_level_and_scripts > includes > level1 > level1.sql > ...

1 select 'my level1 query' as script_name from dual;

2 @@level2/level2.sql

level5.sql U X

show_level_and_scripts > includes > level1 > level2 > level3 > level4 > level5 > level5.sql > ...

1 show scripts

2 show level

3 select 'my level5 query' as script_name from dual;

Show level & scripts

<<time>>

show_level_and_scripts

includes

level1

level2

level3

level4

level5

level5.sql

level4.sql

level3.sql

level2.sql

level1.sql

demo.md

main.sql

README.md

PROBLEMS

OUTPUT

TERMINAL

CHAT

bash - show_level_and_scripts

DEMO RAFAL@demo rafal > @main.sql;

=== main.sql start ===

scripts

└─ main.sql

depth: 2

current: /Users/rg/git/sandbox/demo_sqlcl/show_level_and_scripts/main.sql

SCRIPT_NAME

my MAIN level query

SCRIPT_NAME

my level1 query

SCRIPT_NAME

my level2 query

SCRIPT_NAME

my level3 query

SCRIPT_NAME

my level4 query

scripts

└─ main.sql

└─ includes/level1/level1.sql

└─ level2/level2.sql

└─ level3/level3.sql

└─ level4/level4.sql

└─ level5/level5.sql

depth: 7

current: /Users/rg/git/sandbox/demo_sqlcl/show_level_and_scripts/includes/level1/level2/level3/level4/level5/level5.sql

SCRIPT_NAME

my level5 query



Show level & scripts

APEX

[subcommand]

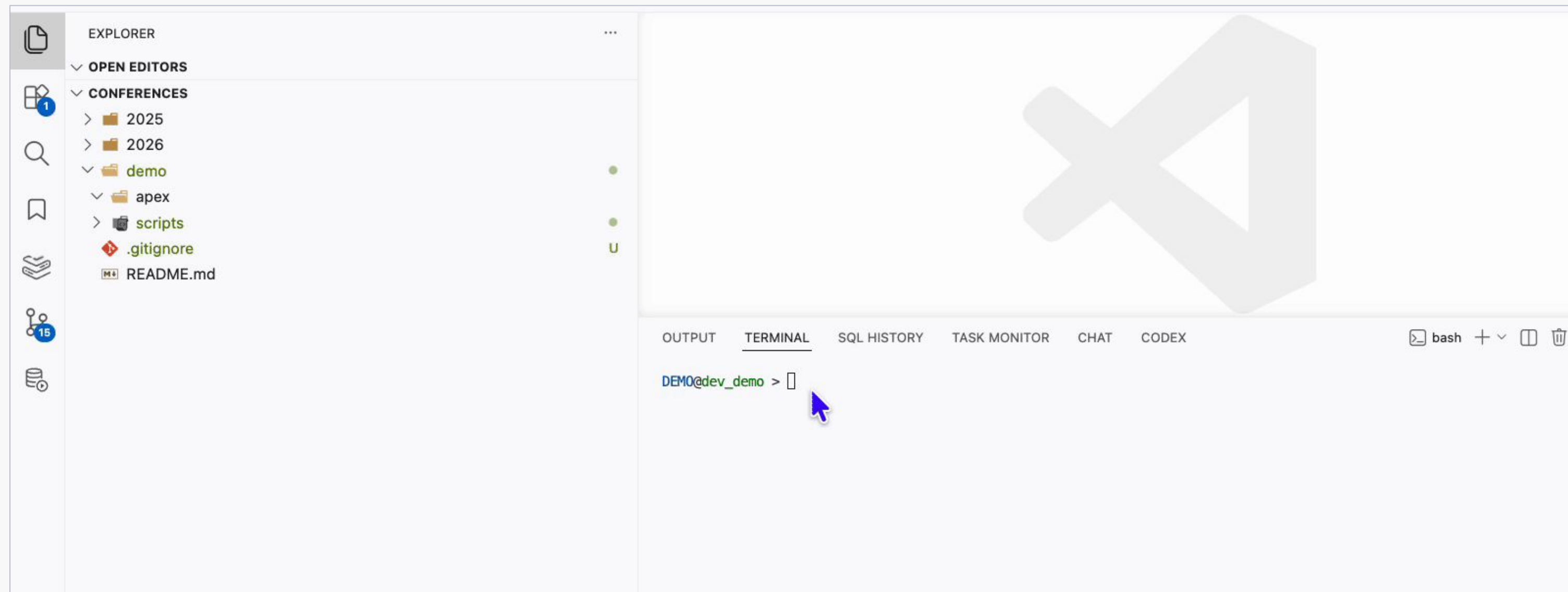
pretius

Questions? Ask here:



Generate sample APEX app (APEXlang)

```
> apex generate -name 'Hello POUG!' -dir demo/apex/poug
```



Validate your APEX app

> apex **validate** -input demo/apex/poug/starter-app

```
DEMO@dev_demo > apex validate -input demo/apex/poug/starter-app
APEXlang Compile Errors:
File: application.apx
Line: 6
Column: 8
Type: INVALID_PROPERTY
Error: Invalid property: imageAlt

File: pages/p00000-global-page.apx
Line: 4
Column: 8
Type: INVALID_PROPERTY
Error: Invalid property: fileUrls
```

It will show you exact location of any issues found in your app



APEX

Validate your APEX app

```
> apex validate -input demo/apex/poug/starter-app
```

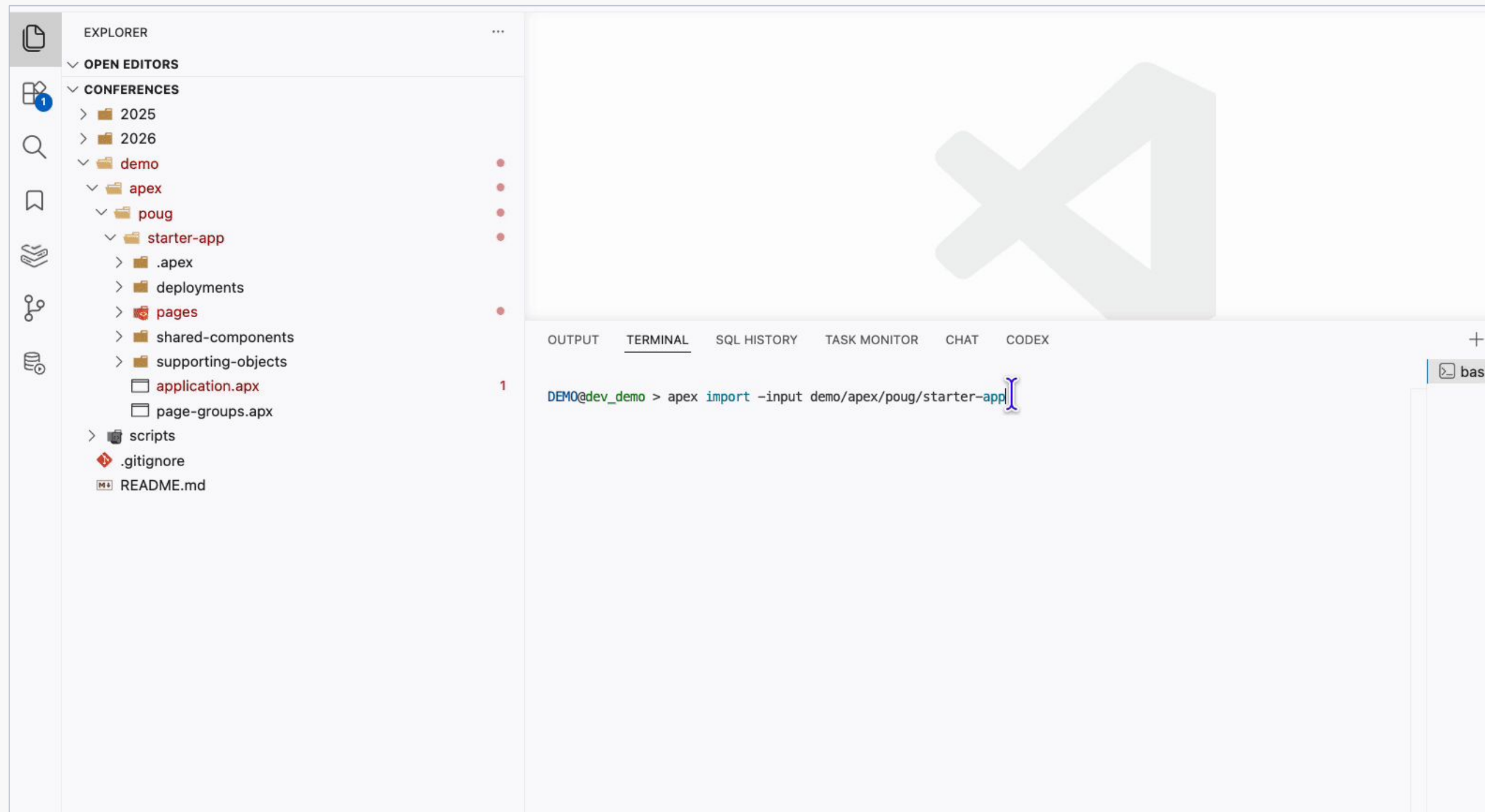
```
DEMO@dev_demo > apex validate -input demo/apex/poug/starter-app  
Validation successful.
```

```
DEMO@dev_demo >
```

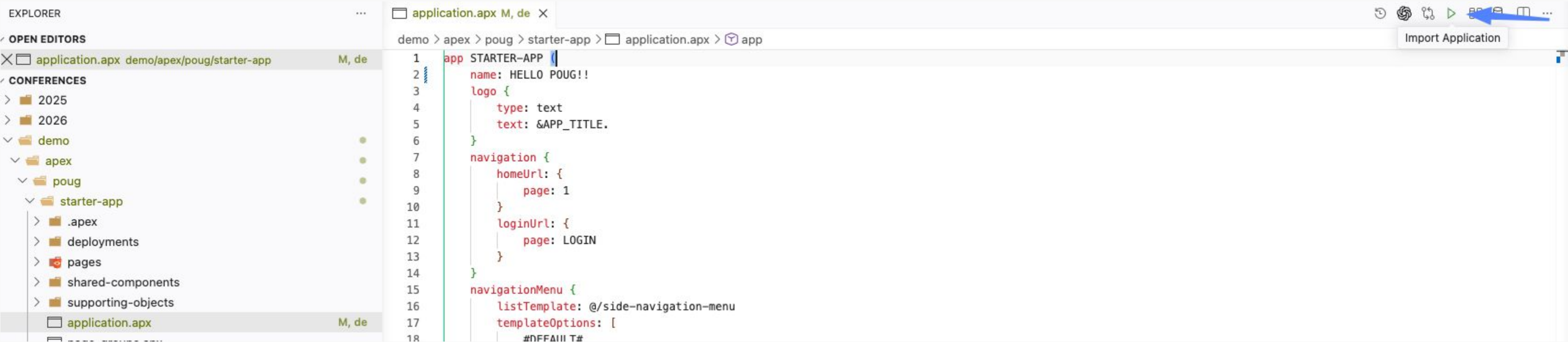


Import application

> apex **import** -input demo/apex/poug/starter-app



Import application directly in VS Code



Go to “help apex” for more

```
DEMO@dev_demo > |
```



SQLcl Project (CI/CD)

pretius

Questions? Ask here:



Project (CI/CD)

- “Project” is an CI/CD extension in SQLcl
- Introduced in 2nd half of 2024
- Contains several sub-commands to help developers manage database & APEX changes
- It uses **Liquibase** & **GIT** in a simple & developer-friendly way



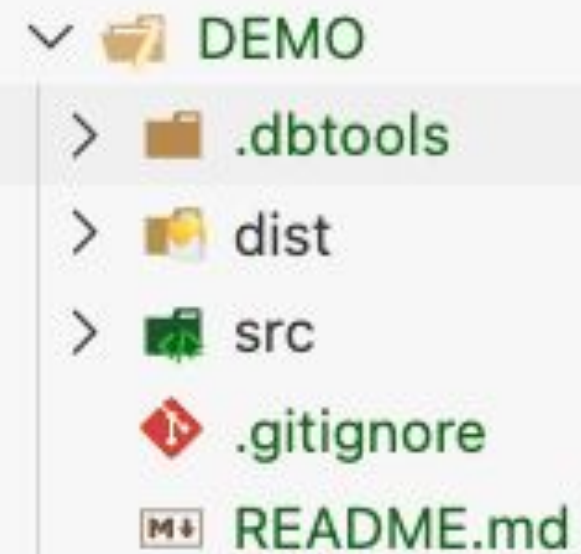
Initialize a new project (GIT repo required)

- > sql -name dev_demo
- > project **init** -name DEMO -schemas DEMO

```
DEMO@dev_demo > project init -name DEMO -schemas DEMO
```

PROJECT DETAILS

```
-----  
Project name:    DEMO  
Schema(s):      DEMO  
Directory:       /Users/rg/DEMO  
Connection name: dev_demo  
Project root:    DEMO  
Your project has been successfully created  
DEMO@dev_demo >
```

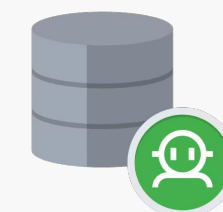
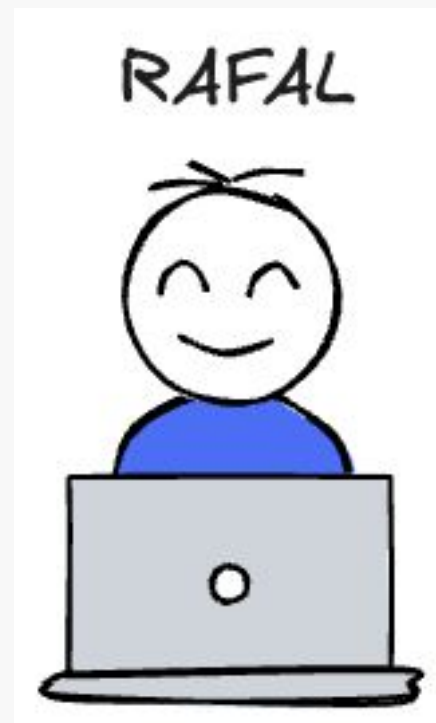


A file explorer view showing the contents of the DEMO project directory. The directory is expanded, showing subdirectories .dbtools, dist, and src. It also shows files .gitignore and README.md. The .gitignore file has a green 'U' icon next to it, and the README.md file has a green 'U' icon next to it.



Project (CI/CD)

Make some Database Objects & APEX changes



Export ALL database & APEX objects

pretius

COUNT(*)	OBJECT_TYPE
142	INDEX
51	TABLE
30	LOB
1	CREDENTIAL
23	TRIGGER
6	PACKAGE
7	VIEW
2	FUNCTION
6	PACKAGE BODY
14	SEQUENCE
2	PROCEDURE

▼	📁 APEX	🔍 ↺
>	📁 APEXToGo (103)	↺
>	📁 Sample Application Search (111)	
>	📁 Sample Document Generator (106)	
>	📁 Sample Interactive Grids (104)	
>	📁 Sample Master Detail (110)	
>	📁 Sample Trees (105)	
>	📁 Sample Vector Search (109)	
>	📁 Universal Theme 24.2 Reference (107)	

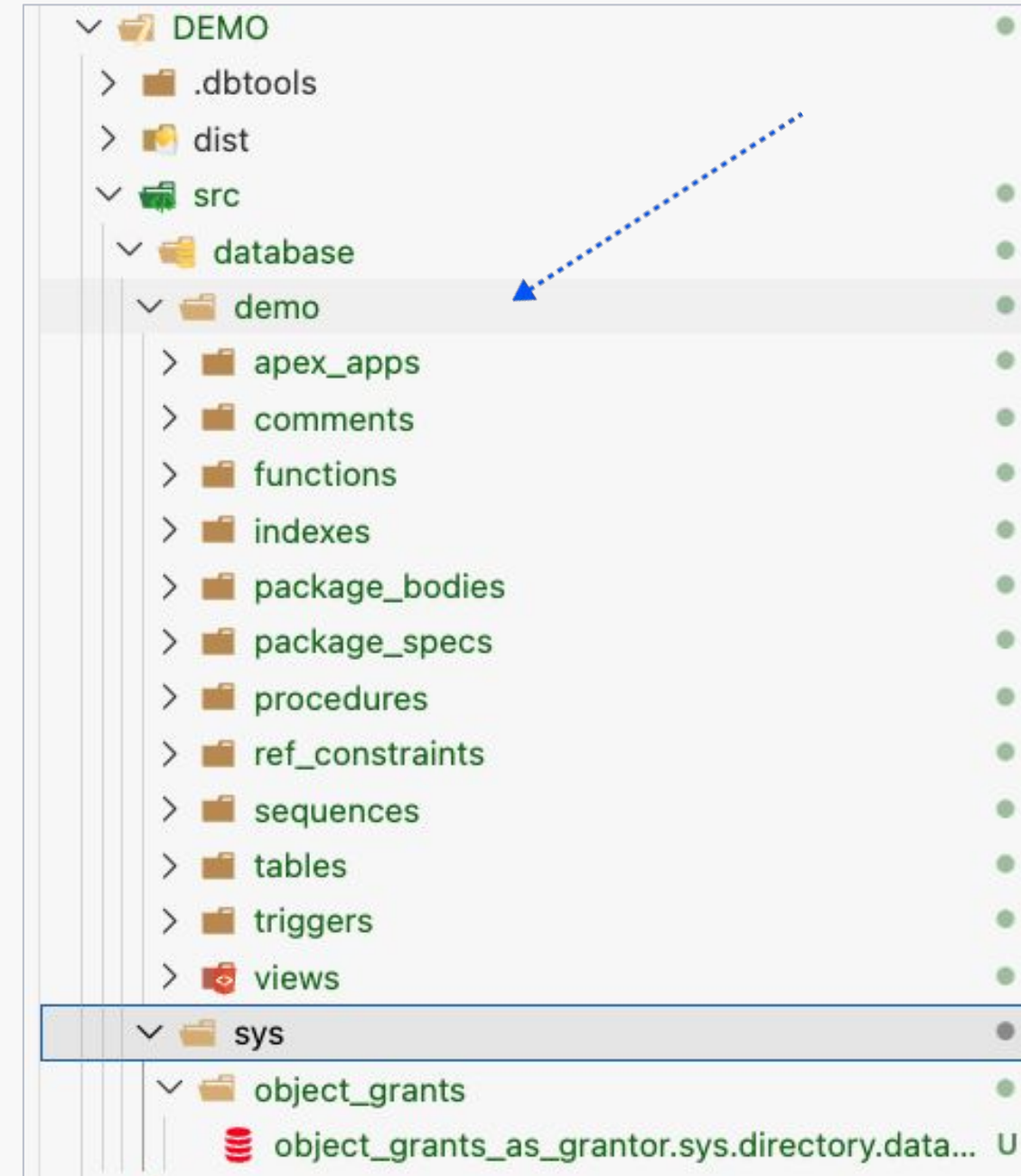


```
> project export
```

```

DEMO@dev_demo > project export
The current connection DEMO will
*** TABLES ***
*** OBJECT_GRANTS ***
*** APEX_APPLICATIONS ***
*** VIEWS ***
*** FUNCTIONS ***
*** PACKAGE_SPECS ***
*** PACKAGE_BODIES ***
*** PROCEDURES ***
*** SEQUENCES ***
*** INDEXES ***
*** TRIGGERS ***
*** REF_CONSTRAINTS ***
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
Exporting Workspace DEMO - appli
-----
APEX_APPLICATION                                8
COMMENT                                          5
FUNCTION                                         2
GRANT                                            2
INDEX                                           61
PACKAGE_BODY                                    6
PACKAGE_SPEC                                    6
PROCEDURE                                       2
REF_CONSTRAINT                                30
SEQUENCE                                        3
TABLE                                           51
TRIGGER                                         23
VIEW                                            7
-----
Exported 206 objects
Elapsed 208 sec

```



Project Stage

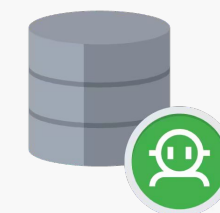
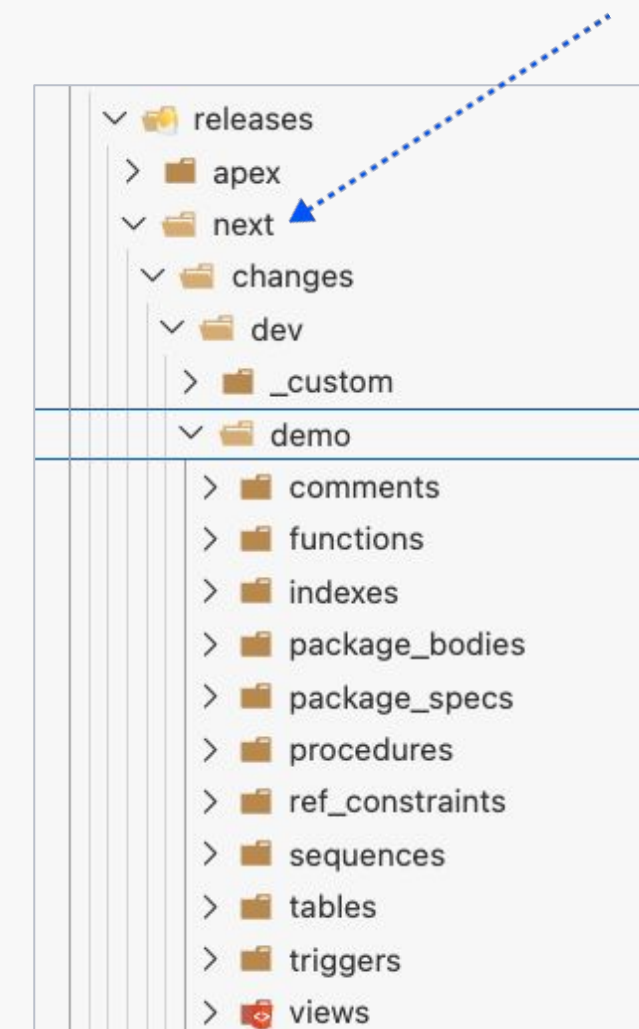
- Compares the current branch (**dev**) with the target branch (**test**) to create Liquibase changelogs that apply the necessary database updates.
- > project **stage** -branch-name test

```
DEMO@dev_demo > project stage -branch-name test
The current connection DEMO will be used for all operations
```

Stage is Comparing:

Old Branch	refs/heads/test
New Branch	refs/heads/dev

Stage processing completed, please review and commit your changes to repository



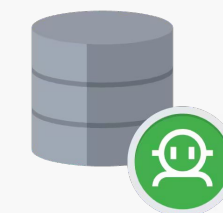
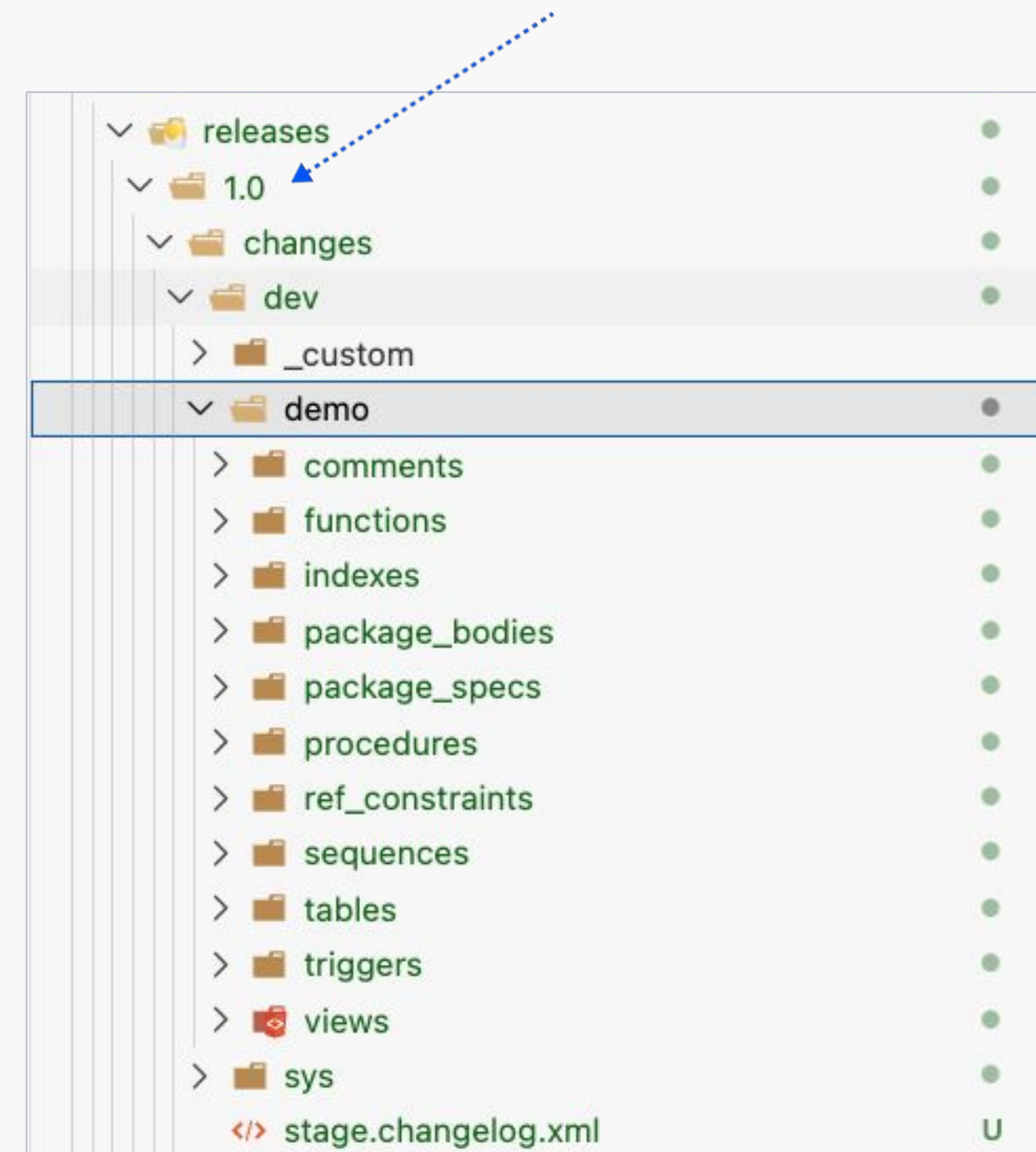
Project (CI/CD)

Project Release

- Transitions current work set into a released state and initiates a fresh set of work
- Once the code is moved into a release, it must not be altered.

> project **release** -version 1.0

```
DEMO@dev_demo > project release -version 1.0  
Process completed successfully  
DEMO@dev_demo > |
```

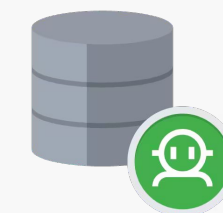


Project (CI/CD)

Liquibase syntax and files are automatically created:

```
eba_demo_dg_orders.sql U X
DEMO > dist > releases > 1.0 > changes > dev > demo > tables > eba_demo_dg_orders.sql > ...
1  -- liquibase formatted sql
2  -- changeset DEMO:1773775410323 stripComments:false logicalFilePath:dev/demo/tables/eba_demo_dg_orders.sql
3  -- sqlcl_snapshot src/database/demo/tables/eba_demo_dg_orders.sql:null:dc1ca737d177f7fef60aaa0dffff9f9bf1b8458e:create
4
5  create table demo.eba_demo_dg_orders (
6      order_id      number generated by default on null as identity not null enable,
7      order_datetime timestamp(6) not null enable,
8      customer_id    number not null enable,
9      order_status   varchar2(10 byte) not null enable
10 );
```

```
stage.changelog.xml U X
DEMO > dist > releases > 1.0 > changes > dev > stage.changelog.xml
1  <?xml version="1.0" encoding="UTF-8"?>
2  <databaseChangelog xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xmlns="http://www.liquibase.org/xml/ns/dbchangelog"
4      xsi:schemaLocation="http://www.liquibase.org/xml/ns/dbchangelog
5          http://www.liquibase.org/xml/ns/dbchangelog/dbchangelog-4.3.xsd">
6      <include file="sys/object_grants/object_grants_as_grantor.sys.directory.data_pump_dir.sql" relativeToChangelogFile="true"/>
7      <include file="demo/sequences/eba_demo_ig_dept_seq.sql" relativeToChangelogFile="true"/>
8      <include file="demo/sequences/eba_demo_ig_emp_seq.sql" relativeToChangelogFile="true"/>
9      <include file="demo/sequences/eba_demo_ig_people_seq.sql" relativeToChangelogFile="true"/>
10     <include file="demo/functions/varchar2_to_blob.sql" relativeToChangelogFile="true"/>
11     <include file="demo/package_specs/eba_demo_ig_text_pkg.sql" relativeToChangelogFile="true"/>
12     <include file="demo/package_specs/eba_demo_md_data_pkg.sql" relativeToChangelogFile="true"/>
13     <include file="demo/package_specs/eba_demo_search_data_pkg.sql" relativeToChangelogFile="true"/>
14     <include file="demo/package_specs/eba_demo_search_text_pkg.sql" relativeToChangelogFile="true"/>
15     <include file="demo/package_specs/eba_sample_vector_search_pkg.sql" relativeToChangelogFile="true"/>
16     <include file="demo/tables/eba_demo_dg_dept.sql" relativeToChangelogFile="true"/>
17     <include file="demo/tables/eba_demo_dg_emp.sql" relativeToChangelogFile="true"/>
18     <include file="demo/tables/eba_demo_ig_dept.sql" relativeToChangelogFile="true"/>
```

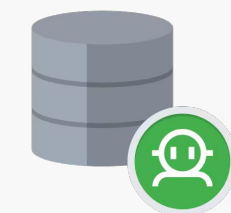
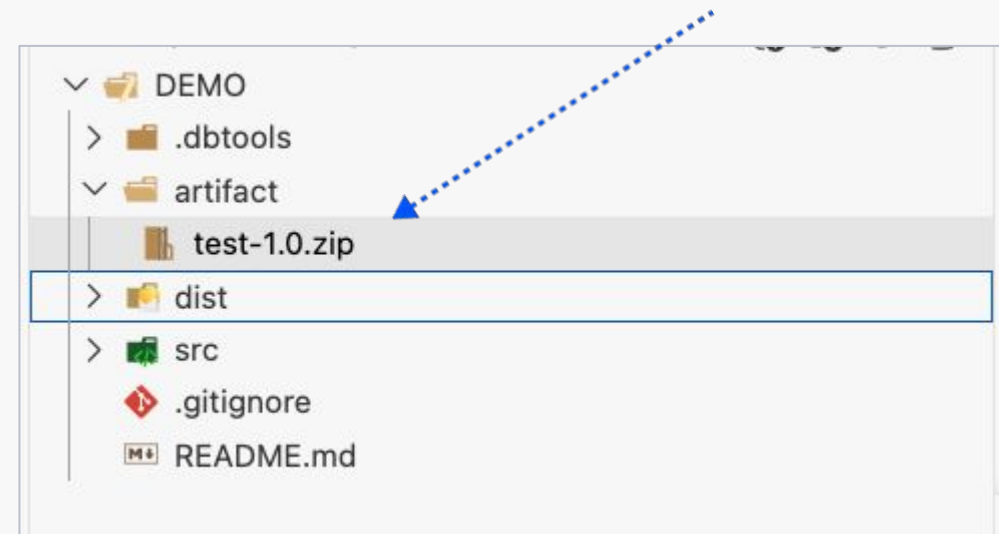


Project Gen-Artifact

- Generates an artifact representing the current state of your project.

> project **gen-artifact** -name test -version 1.0

```
DEMO@dev_demo > project gen-artifact -name test -version 1.0  
Your artifact has been generated test-1.0.zip  
DEMO@dev_demo >
```

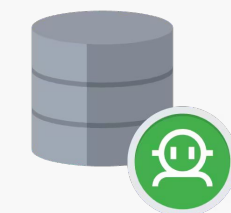


Project Deploy

- Deploys the artifact on a target database (TEST)

> project **deploy** -file artifact/test-1.0.zip

```
ADMIN@test_admin > project deploy -file artifact/test-1.0.zip
Starting the migration...
|
```



How Liquibase in SQLcl works?

- Liquibase tracks which database changes has been applied to which environment
- Keeps your database structure consistent across dev, test,prod etc.
- Changes are tracked in DATABASECHANGELOG, and executed SQL can be found in DATABASECHANGELOG_ACTIONS

```
DEMO@test_demo > select id, filename, dateexecuted, comments, md5sum from databasechangelog;
```



Commands summary

- ❑ **Initialize** a new project

```
project init -name bombshell -schemas HR,CO
```
- ❑ **Export** all DB objects
 - ❑ Export some objects

```
HR.COUNTRIES, APEX.101
```
 - ❑ Add custom file/script

```
project stage add-custom -file-name my_file
```
- ❑ **Stage** (gen. changelogs/changesets)

```
project stage -branch-name test
```
- ❑ **Release**

```
project release -version 1.1
```
- ❑ **Generate** artifact (zip)

```
project gen-artifact -name test -version 1.1
```
- ❑ **Deploy**

```
project deploy -file artifact/test-1.1.zip
```



DIFF – compare two schemas and synchronize them

Questions? Ask here:



DIFF

- Use it to compare two Oracle DB schemas and generate SQL scripts (artifact) with necessary changes to synchronize them.
- Extremely helpful if you struggle with database schema drift (environments are not equal).
- Use DIFF before you start using CI/CD Tools (SQLcl Projects, Liquibase etc.).

More about DIFF at
Tomas Kucharzyk's
blog [here](#):



[DIFF documentation](#)



Diff

SOURCE

TARGET

DEV [schema DEMO]

- 12 database objects

TEST [schema DEMO]

- 11 database objects

> **diff -source** dev_demo **-target** test_demo

```
@ > diff -source dev_demo -target test_demo
[diff - info] Preparing clean project directory: /var/folders/dj/nwvnq0jn5992_8rxcbcv5xbr0000gn/T/diff_project_560707579396177596
[diff - info] Initializing project with schemas: DEMO,TEST
[diff - info] Generating diffs via project stage against: base_DEMO
[diff - info] Generating artifact (SQLcl default location)
[diff - info] Artifact copied from "/var/folders/dj/nwvnq0jn5992_8rxcbcv5xbr0000gn/T/diff_project_560707579396177596/artifact/diff DEMO vs TEST.zip" to "/Users/rg/git/sandbox/demo_sqlcl/diff/diff_DEMO_vs_TEST-1779711258209.zip"
[diff - info] Success. Diffs under dist/releases/next. Logs at: null
@ > |
```



Diff

Review generated SQL scripts

- Unzip and go to releases/next/changes/target_test and review generated files

The screenshot shows an IDE interface. On the left, the EXPLORER pane displays a file tree for a project named DEMO_SQLCL. The tree structure is as follows:

- DEMO_SQLCL
 - apex
 - diff
 - diff_DEMO_vs_TEST-1779711258209
 - env
 - releases
 - next
 - changes
 - target_test
 - demo (selected)
- stage.changelog.sql
- release.changelog.sql
- main.changelog.sql
- utils
- install.sql

On the right, the editor pane shows a diff view of the file `stage.changelog.sql`. The diff is titled `diff > diff_DEMO_vs_TEST-1779711258209 > releases > next > changes > targ`. The content of the diff is as follows:

```
1  --stage
2  @@demo/triggers/loki_before_ddl_tgr.sql
3  @@demo/procedures/eba_demo_ir_data.sql
4  @@demo/functions/perf_sample_get_employee_status.sql
5  @@demo/package_specs/sqlcl_lb_capture.sql
6  @@demo/package_specs/eba_demo_ir_pkg.sql
7  @@demo/comments/esql_1220.sql
8  @@demo/tables/esql_1220.sql
9  @@demo/tables/eba_demo_ir_projects.sql
10 @@demo/tables/eba_demo_ir_emp.sql
11 @@demo/tables/eba_demo_ir_dept.sql
12 @@demo/tables/dbtools$mcp_log.sql
13
```

At the bottom of the IDE, there is a panel with tabs for PROBLEMS, OUTPUT, TERMINAL, and CHAT. The TERMINAL tab is active, showing a prompt `@ >` .



Diff

Review generated SQL scripts

EXPLORER

> OPEN EDITORS

DEMO_SQLCL

apex

diff

diff_DEMO_vs_TEST-1779711258209

env

releases

next

changes

target_test

demo

comments

functions

package_specs

procedures

tables

triggers

stage.changelog.sql

release.changelog.sql

main.changelog.sql

utils

install.sql

stage.changelog.sql

diff > diff_DEMO_vs_TEST-1779711258209 > releases > next > changes > targ

1 --stage

2 @@demo/triggers/loki_before_ddl_tgr.sql

3 @@demo/procedures/eba_demo_ir_data.sql

4 @@demo/functions/perf_sample_get_employee_status.sql

5 @@demo/package_specs/sqlcl_lb_capture.sql

6 @@demo/package_specs/eba_demo_ir_pkg.sql

7 @@demo/comments/esql_1220.sql

8 @@demo/tables/esql_1220.sql

9 @@demo/tables/eba_demo_ir_projects.sql

10 @@demo/tables/eba_demo_ir_emp.sql

11 @@demo/tables/eba_demo_ir_dept.sql

12 @@demo/tables/dbtools\$mcp_log.sql

13

PROBLEMS

OUTPUT

TERMINAL

CHAT

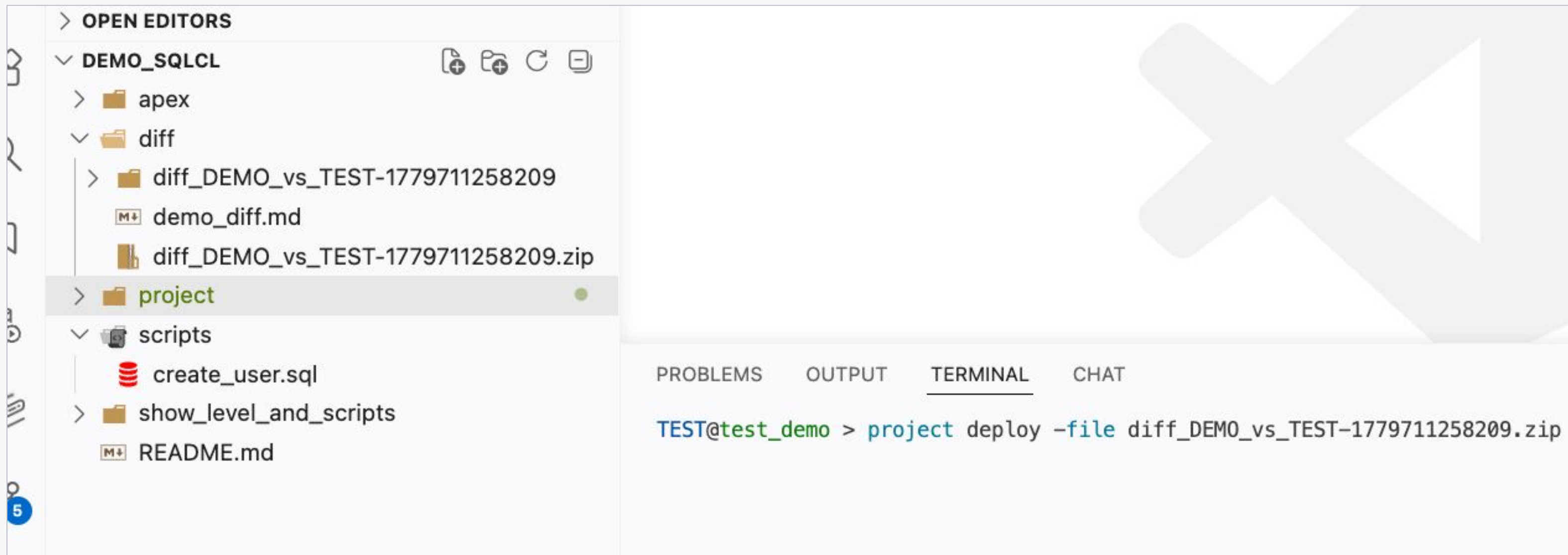
@ >



Diff

Deploy to target schema

- > sql -name test_demo
- > **project deploy** -file [your_filename].zip



Diff

SQLcl Codescan

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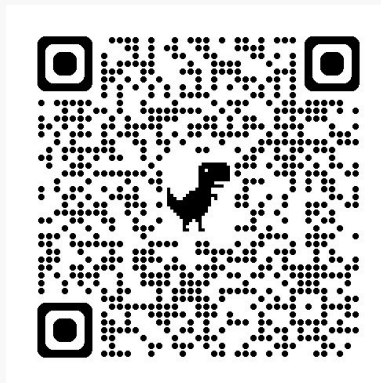
Questions? Ask here:



What is CODESCAN?

- Checks quality of written PL/SQL and SQL
- Follows the rules from Trivadis PL/SQL & SQL Coding Guidelines
- Identifies performance issues stated in Oracle documentation

My blog about
CODESCAN [here:](#)



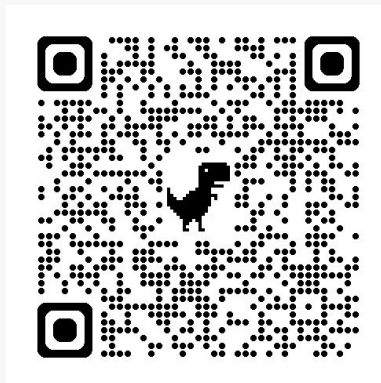
Codescan

Rules checked

```
> codescan -list-rules
```

```
@ > |
```

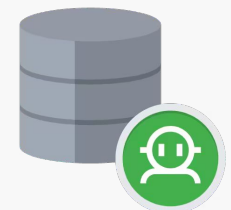
My blog about
CODESCAN [here:](#)



[SQL Performance Issues](#)

[Trivadis PL/SQL & SQL Coding Guidelines](#)

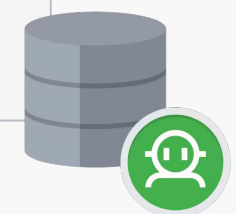
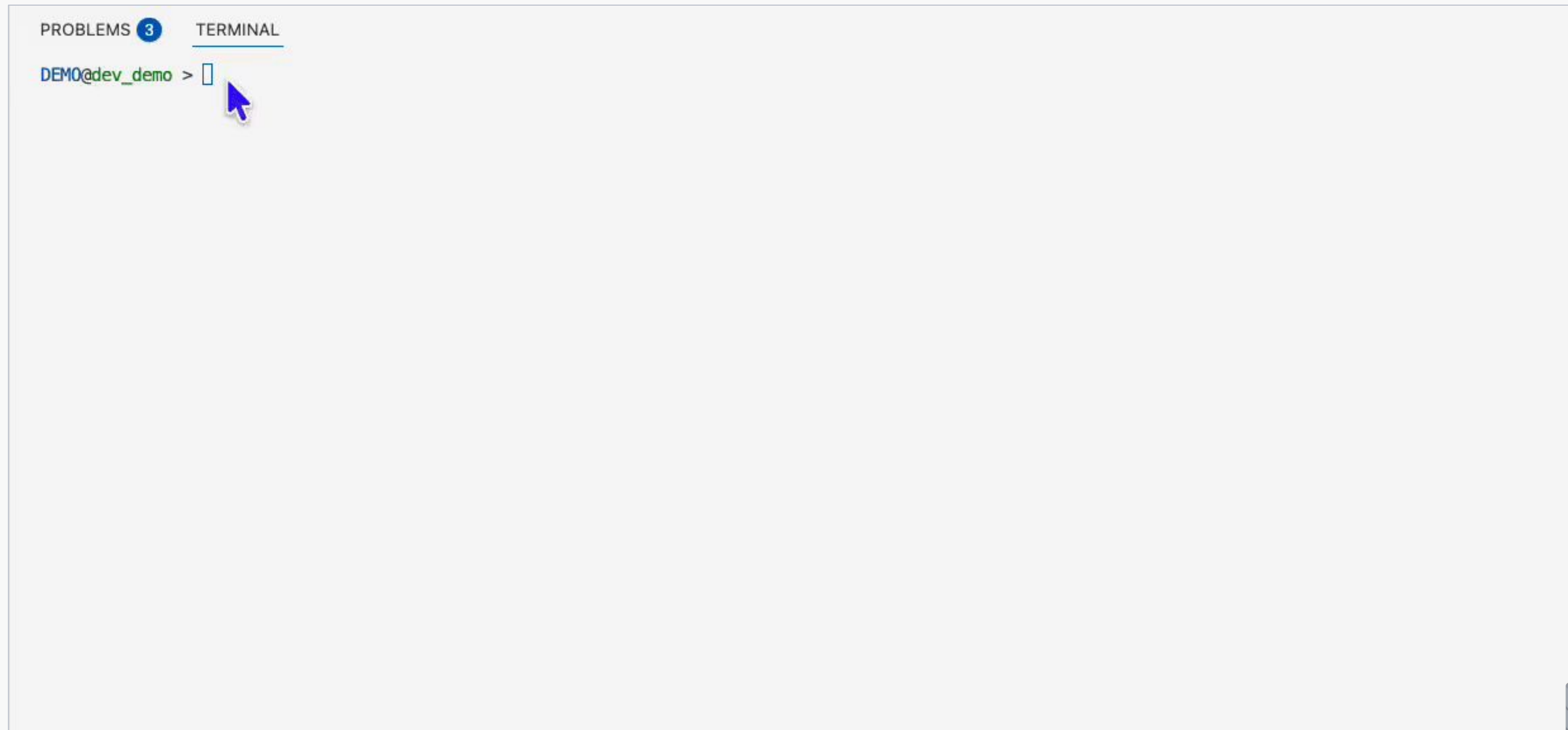
[dbLint rules](#)



Codescan

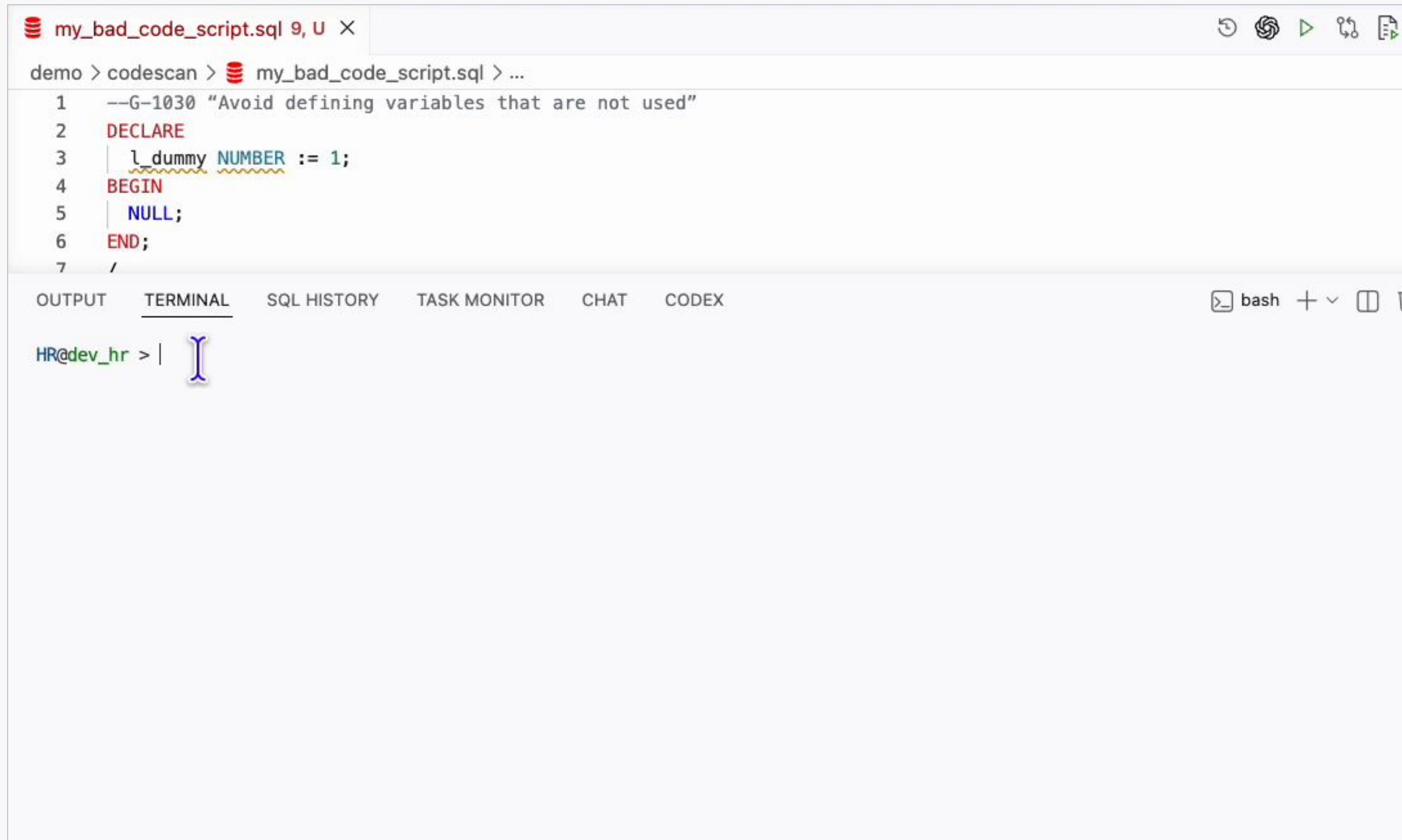
LIVE quality scan

> set **codescan** on [default off]



Codescan

LIVE scan of executed scripts



The screenshot displays the Codescan IDE interface. At the top, a tab is labeled 'my_bad_code_script.sql 9, U'. Below the tab, the breadcrumb path reads 'demo > codescan > my_bad_code_script.sql > ...'. The main editor area contains a SQL script with the following lines:

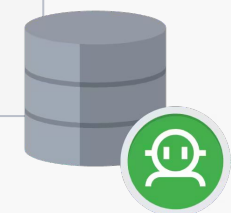
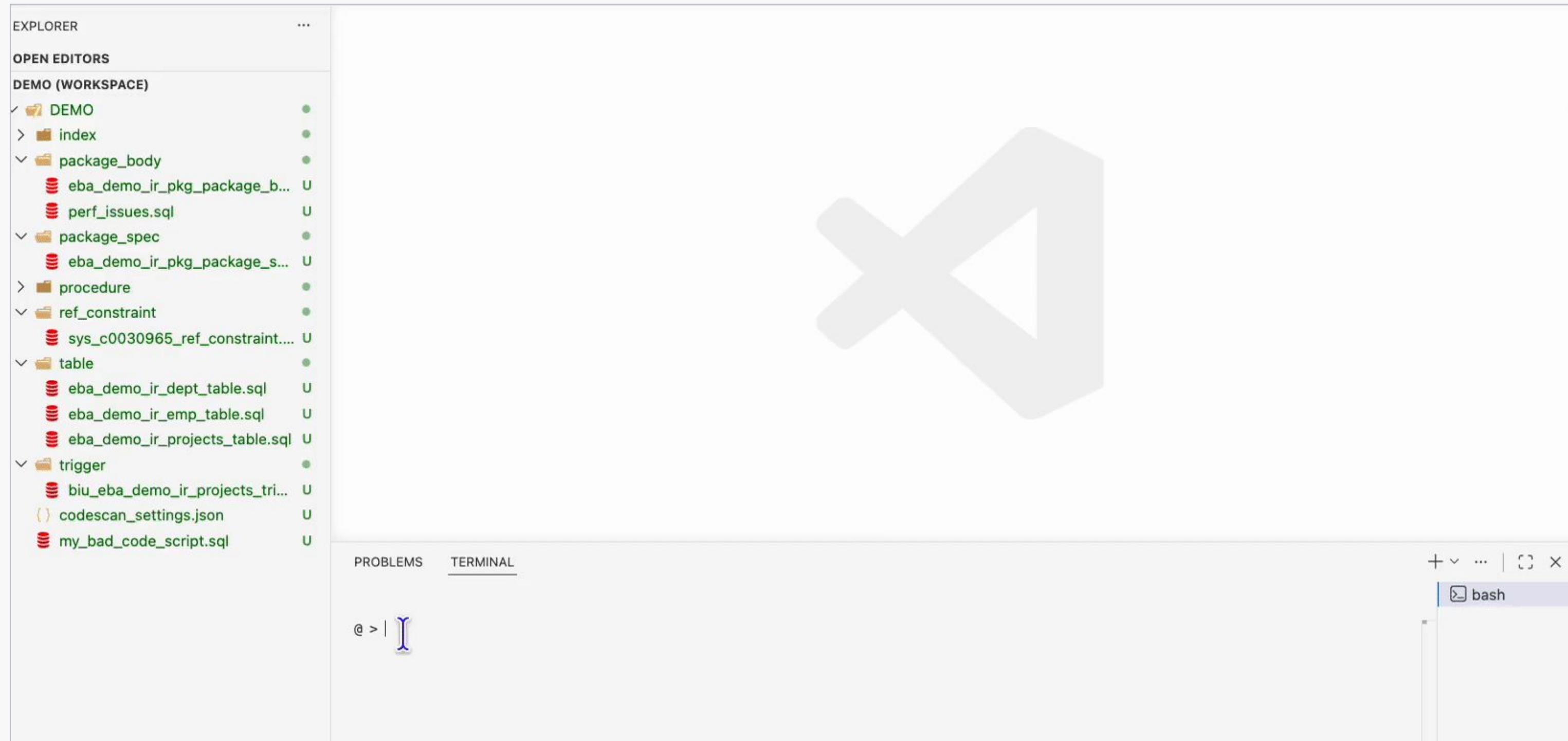
```
1  --G-1030 "Avoid defining variables that are not used"
2  DECLARE
3  |   l_dummy NUMBER := 1;
4  BEGIN
5  |   NULL;
6  END;
7  /
```

The variable 'l_dummy' is underlined with a yellow wavy line, indicating a linting error. Below the editor, a horizontal menu includes 'OUTPUT', 'TERMINAL' (which is selected), 'SQL HISTORY', 'TASK MONITOR', 'CHAT', and 'CODEX'. The 'TERMINAL' pane shows a prompt 'HR@dev_hr > |' with a blue cursor. On the right side of the terminal pane, there is a 'bash' icon and a '+ v' button.



Scanning multiple files (connection not required!)

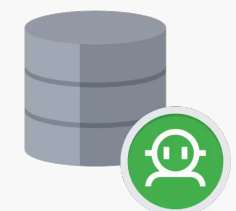
```
> codescan -format json -output my_default_scan.json
```



Ignore rules you don't like / don't agree with

```
{ } my_default_scan.json U X
DEMO > { } my_default_scan.json > { } 0 > [ ] issues
1  [
2  {"file": "/Users/rg/DEMO/my_bad_code_script.sql", "issues": [
3    { "line": 2, "col": 2, "ruleNo": "G-1030", "msg": "Avoid defining variables that are not used"},
4    { "line": 8, "col": 7, "ruleNo": "G-3145", "msg": "Avoid using SELECT * directly from a table or view"},
5    { "line": 14, "col": 18, "ruleNo": "G-7410", "msg": "Avoid standalone functions put your functions in packages"},
6    { "line": 14, "col": 59, "ruleNo": "G-7160", "msg": "Always explicitly state parameter mode"},
7    { "line": 24, "col": 1, "ruleNo": "G-3130", "msg": "Try to use ANSI SQL-92 join syntax"},
8    { "line": 25, "col": 18, "ruleNo": "PSR-203", "msg": "An inline view or table detected that was OUTER-joined on optional side of the join"},
9    { "line": 34, "col": 1, "ruleNo": "G-5060", "msg": "Avoid unhandled exceptions"},
10   { "line": 36, "col": 9, "ruleNo": "PSR-102", "msg": "Scalar subqueries are subqueries in SELECT list. They must return exactly one value"},
11   { "line": 42, "col": 1, "ruleNo": "G-5010", "msg": "Try to use a error/logging framework for your application"},
12   { "line": 42, "col": 1, "ruleNo": "G-7510", "msg": "Always prefix Oracle supplied packages with owner schema name"}
13 ]},
14 {"file": "/Users/rg/DEMO/trigger/biu_eba_demo_ir_projects_trigger.sql", "issues": [
15   { "line": 2, "col": 7, "ruleNo": "G-7730", "msg": "Avoid multiple DML events per trigger"},
16   { "line": 6, "col": 12, "ruleNo": "G-9030", "msg": "Try to define a default value on conversion errors"},
17   { "line": 6, "col": 69, "ruleNo": "G-5060", "msg": "Avoid unhandled exceptions"},
18   { "line": 14, "col": 32, "ruleNo": "G-5050", "msg": "Avoid use of the RAISE_APPLICATION_ERROR built-in procedure with a hard-coded 20nnn"}
19 ]},
20 {"file": "/Users/rg/DEMO/package_body/eba_demo_ir_pkg_package_body.sql", "issues": [
21   { "line": 14, "col": 34, "ruleNo": "G-2160", "msg": "Avoid initializing variables using functions in the declaration section"},
22   { "line": 21, "col": 41, "ruleNo": "G-1050", "msg": "Avoid using literals in your code"},
23   { "line": 22, "col": 41, "ruleNo": "G-1050", "msg": "Avoid using literals in your code"},
24 ]}
```

* Ignore doesn't work with Performance Rules (PSR-XXX)



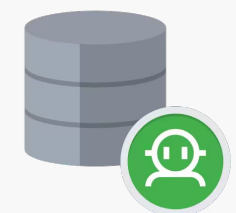
Ignore rules you don't like / don't agree with

- Simply define .json file with ignored rules



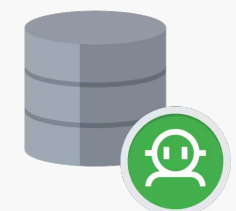
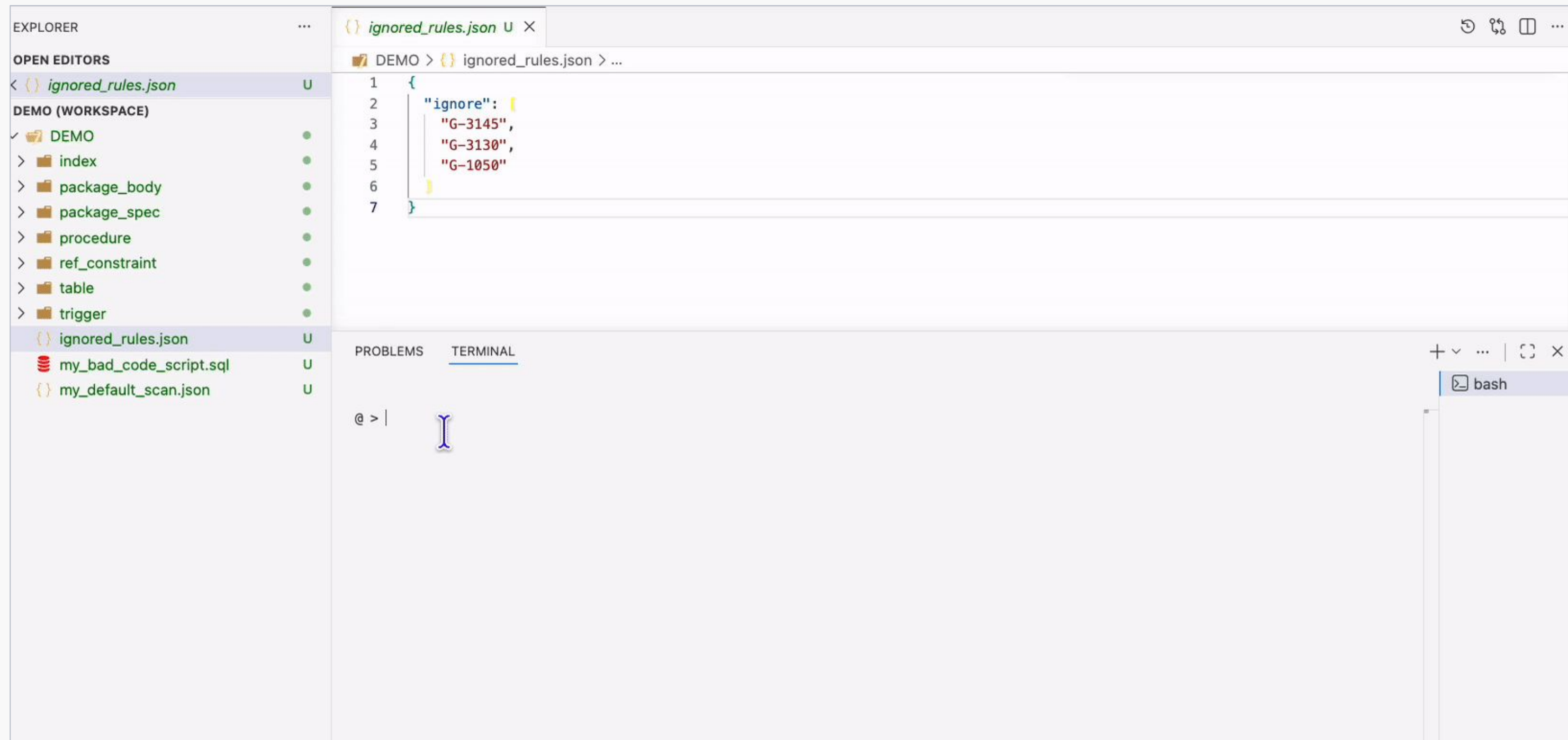
```
{
  "ignore": [
    "G-3145",
    "G-3130",
    "G-1050"
  ]
}
```

* Ignore doesn't work with Performance Rules (PSR-XXX)



Ignore rules you don't like / don't agree with

> **codescan** -format json -output custom_scan.json -**settings** ignored_rules.json



SQLcI MCP + AI

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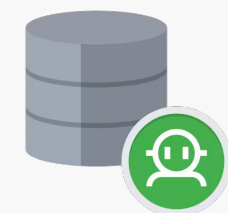
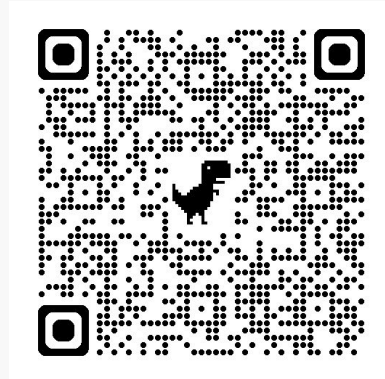
Questions? Ask here:



What is MCP?

- **Model Context Protocol** is a standard that allows AI models to connect with external data e.g databases
- Think of it as USB-C port between your AI and database.

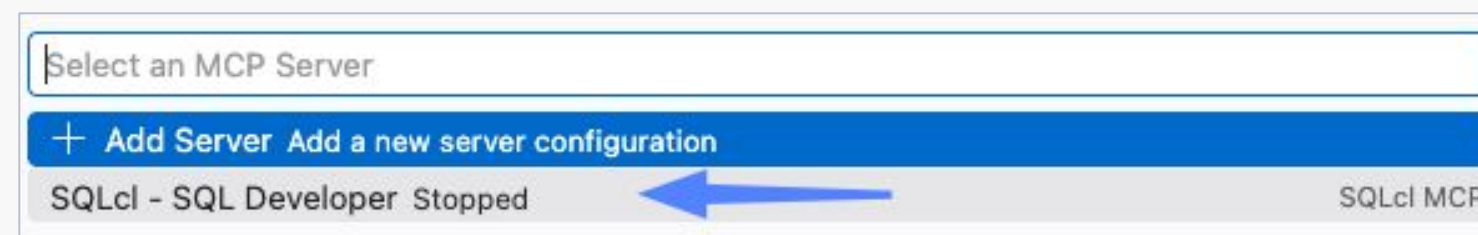
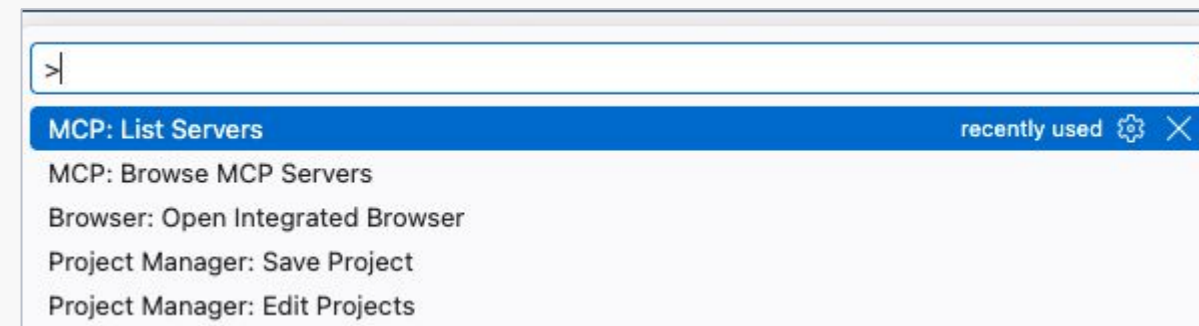
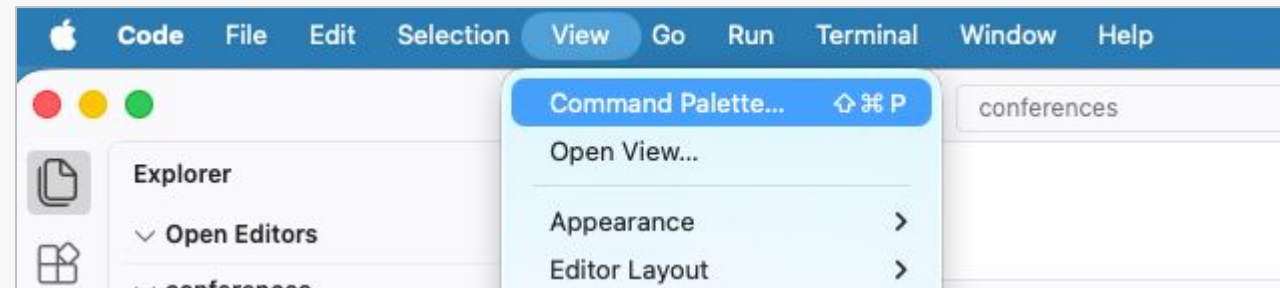
My intro blog
about MCP [here:](#)



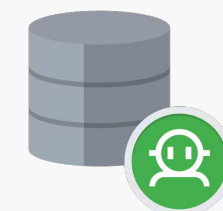
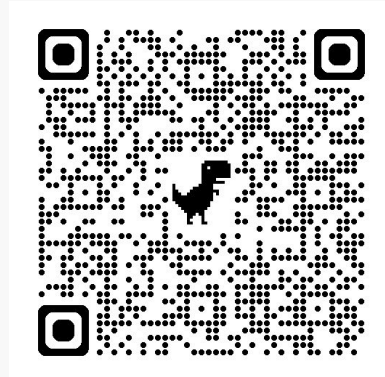
SQLcl + MCP AI

You already have SQLcl MCP in VS Code

- MCP "SQLCI - SQL Developer" is built-in (if you have SQL Developer Extension)

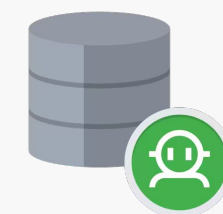
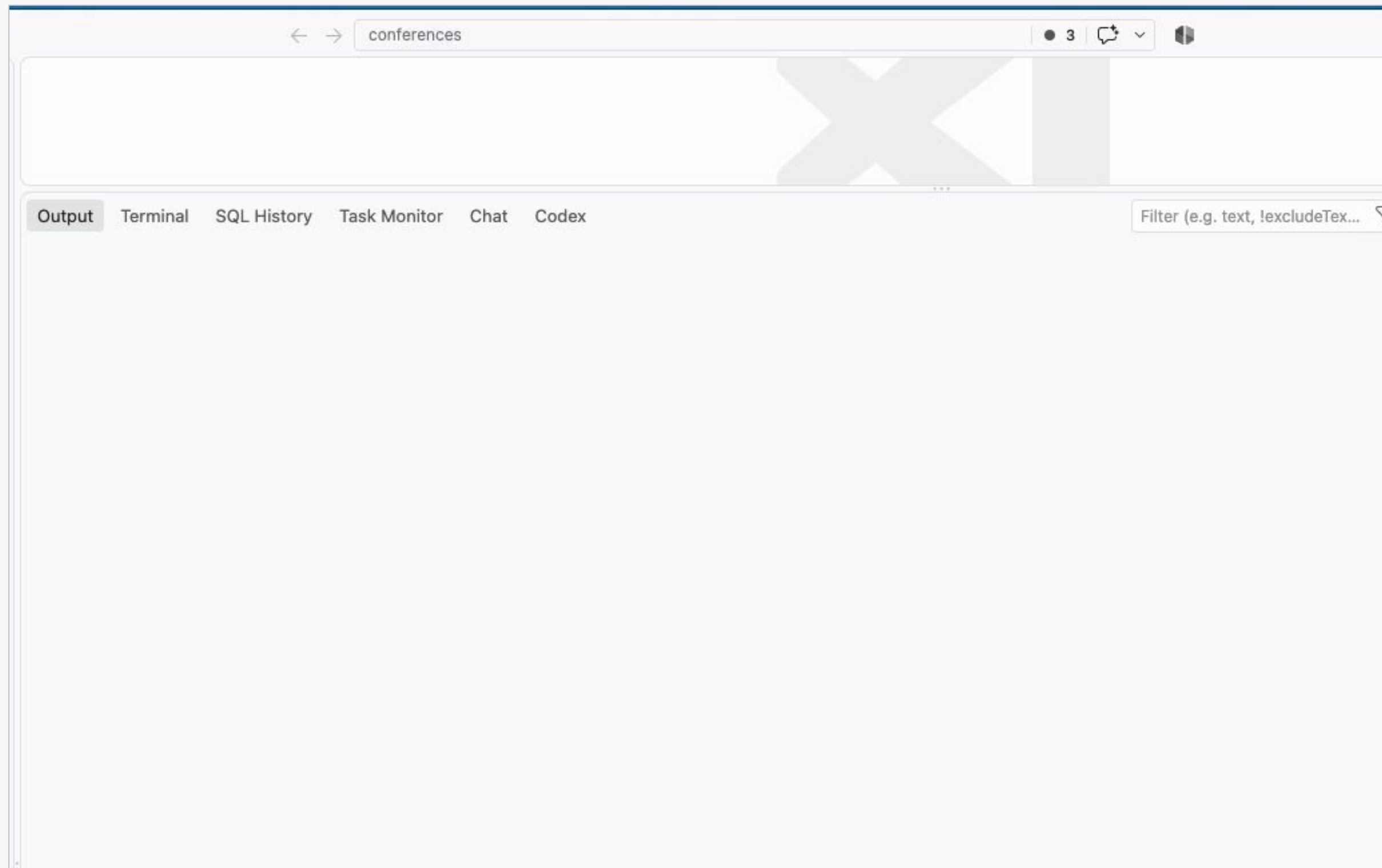


My intro blog
about MCP [here:](#)

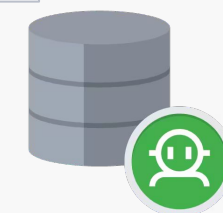
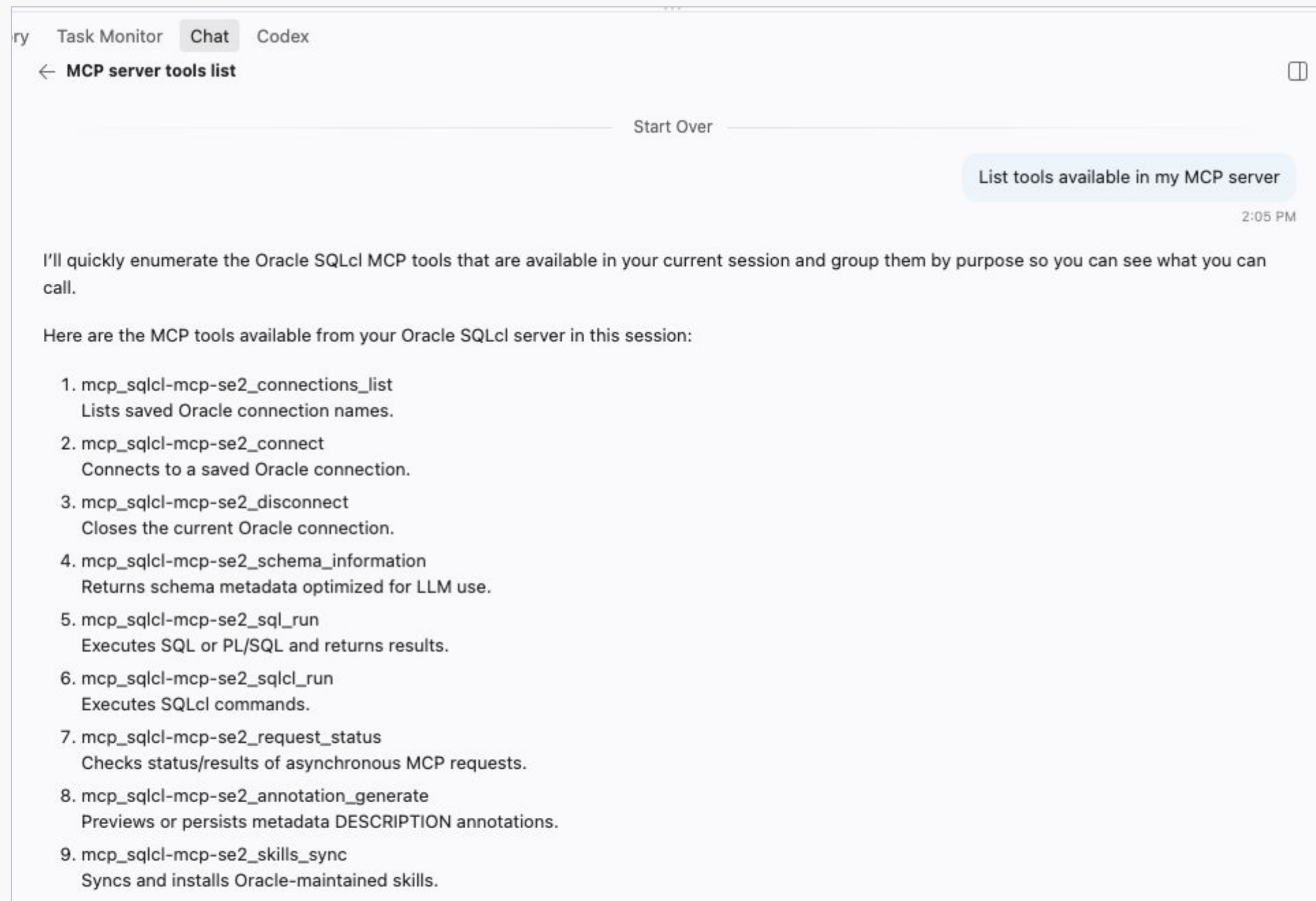


SQLcl + MCP AI

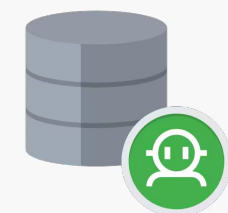
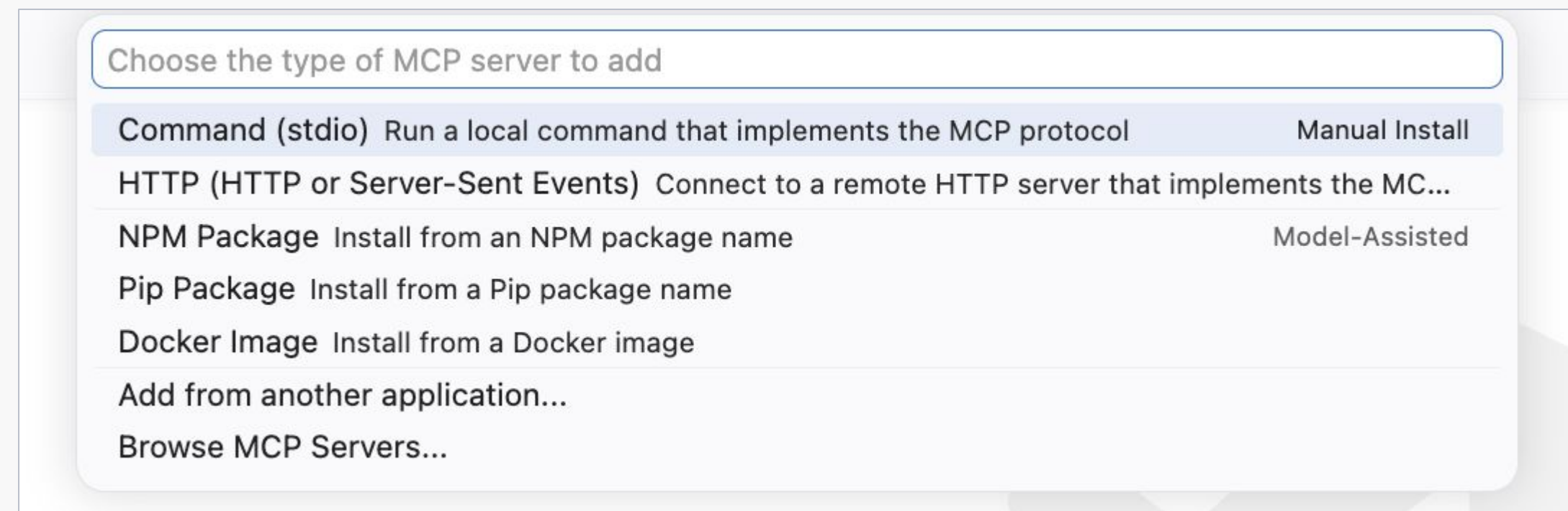
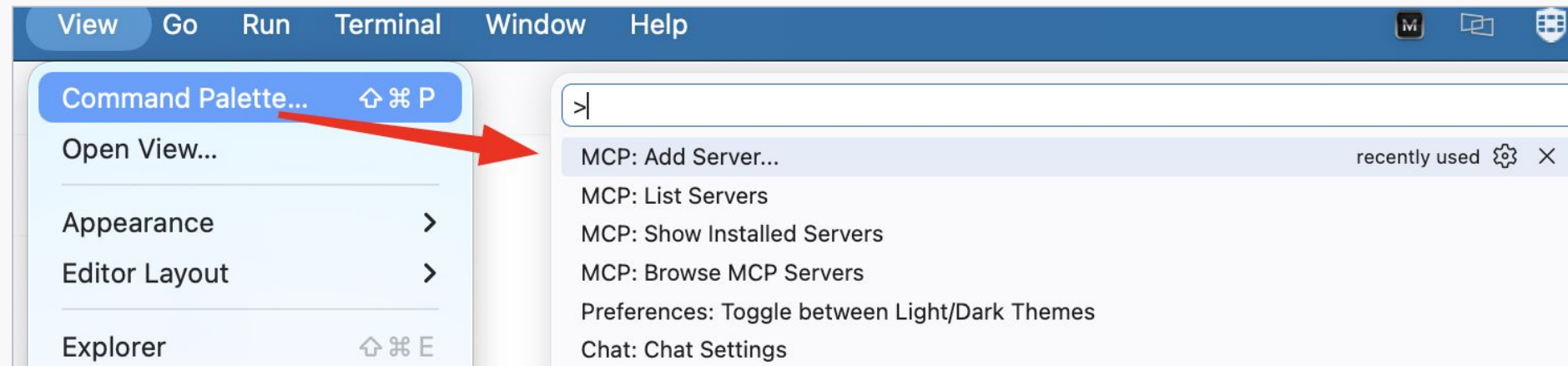
You already have SQLcl MCP in VS Code



You already have SQLcl MCP in VS Code



Add your own SQLcl MCP in VS Code



Add MCP in VS Code (for Github Copilot)

Enter Command

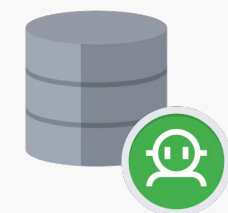
/Users/rg/apps/oracle/sqlcl/bin/sql|

Press 'Enter' to confirm your input or 'Escape' to cancel

Enter Server ID

POUG_MCP

Press 'Enter' to confirm your input or 'Escape' to cancel



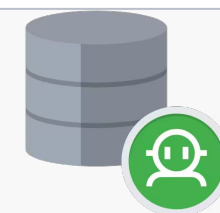
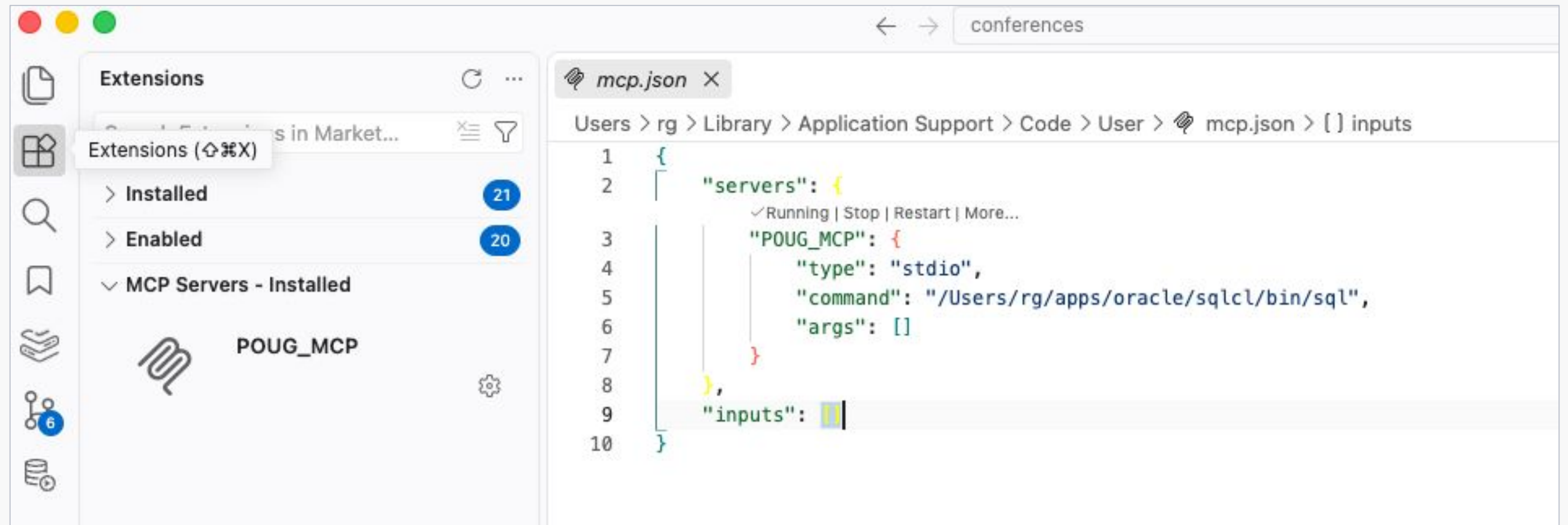
Add MCP in VS Code (for Github Copilot)

Add MCP Server

Select the configuration target

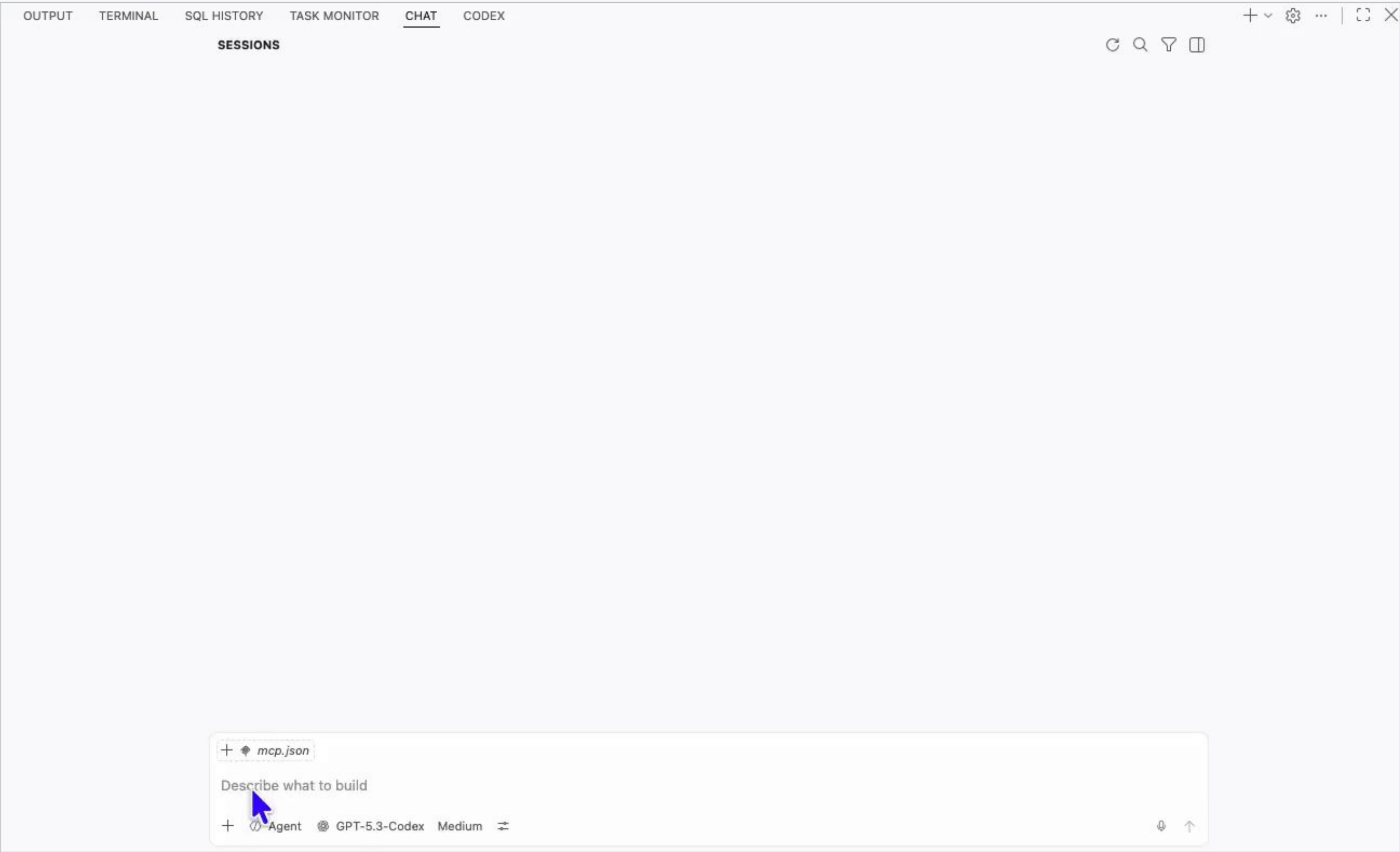
Global Available in all workspaces, runs locally

Workspace Available in this workspace, runs locally



SQLcl + MCP AI

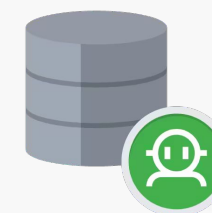
Now, AI has access to your DB



SQLcl + MCP AI

MCP and security?

- Assign minimum permissions
 - create db user account with minimum permissions required for its tasks
- Avoid production database access
 - use sanitized data, read-only copies or dedicated data subsets
- Audit LLM activity
 - auto-created DBTOOLS\$MCP_LOG table for activity logging



Configure restrict levels for SQLcl MCP Server

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Level	Access	
4 (default)	Most restrictive	Blocks all commands from level 1-4
3	Highly restrictive	Disable running external scripts in SQLcl (@, @@)
2	Medium restrictive	Prevent saving SQL results to external files (SAVE, SPOOL, STORE)
1	Minimally restrictive	Prevents HOST or EDIT commands
0	Unrestricted	Allows all commands

[Commands Restricted by Level](#)

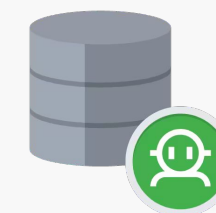
[Configuring Restrict Levels](#)



Security SQLcl + MCP AI

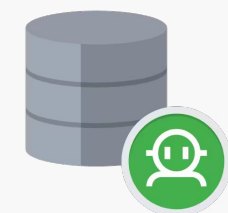
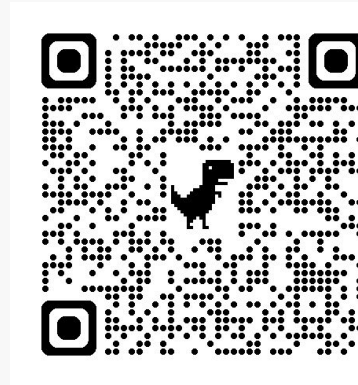
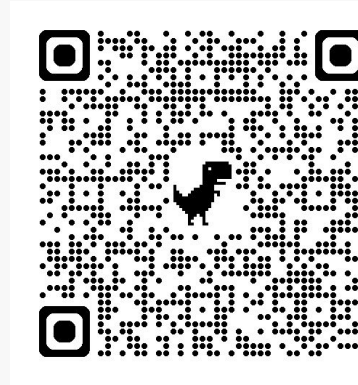
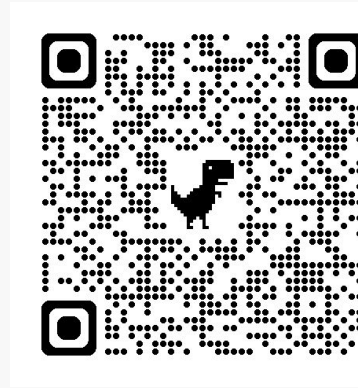
Configure restrict levels for SQLcl MCP Server

```
mcp.json U X
.vscode > mcp.json > ...
1 {
2   "inputs": [],
3   "servers": {
4     ✓Running | Stop | Restart | 7 tools | 2 prompts | More...
5     "conference_mcp": {
6       "type": "stdio",
7       "command": "/Users/rq/apps/oracle/sqlcl/bin/sql",
8       "args": ["-R", "1", "-mcp"]
9     }
10  }
11 }
```



Worth reading:

- [Getting Started with our MCP Server for Oracle Database \(by Jeff Smith\)](#)
- [SQLcl MCP Server Official Documentation](#)
- [Least Privilege Isn't Optional When an Agent Is Doing the Database Work.](#)



Install AI DBTools Skills via SQLcl

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Questions? Ask here:



List existing AI skills

```
> sql /nolog  
> skills list
```

```
@ > skills list -v  
INFO: Listing DBTools skills from /Users/rg/.dbtools/skills/repositories/oracle-skills.  
WARN: No local DBTools skills repository found. Run skills sync to synchronize it.  
@ > |
```



Add or update skills

> **skills sync**

```
@ > skills sync
SUCCESS: Synchronized DBTools skills repository to /Users/rg/.dbtools/skills/repositories/oracle-skills version latest (refs/heads/main -> 11e528de5ea0686b53adf1d853c909a4bf9ed4d9).
SUCCESS: Installed 15 entries into agent skill locations.
WARN: Skipped existing skill entries: /Users/rg/.copilot/skills/apex, /Users/rg/.copilot/skills/db, /Users/rg/.copilot/skills/fusion, /Users/rg/.copilot/skills/graal, /Users/rg/.copilot/skills/oci
SUCCESS: DBTools skills sync log: /Users/rg/.dbtools/skills/logs/skills-sync-20260525T104313.345Z.log
@ >
```

```
@ > skills list -v
INFO: Listing DBTools skills from /Users/rg/.dbtools/skills/repositories/oracle-skills.
```

+ Skill	+ Status	+ Agents
apex	INSTALLED	cline, dexto, github-copilot, warp
db	INSTALLED	cline, dexto, github-copilot, warp
fusion	INSTALLED	cline, dexto, github-copilot, warp
graal	INSTALLED	cline, dexto, github-copilot, warp
oci	INSTALLED	cline, dexto, github-copilot, warp



AI Skills

```
HR@dev_hr > skills list
```

Skill	Status	Agents
apex	INSTALLED	cline, codex, dexto, github-copilot, warp
db	INSTALLED	cline, codex, dexto, github-copilot, warp
fusion	INSTALLED	cline, codex, dexto, github-copilot, warp
<u>graal</u>	INSTALLED	cline, codex, dexto, github-copilot, warp
oci	INSTALLED	cline, codex, dexto, github-copilot, warp



Customise SQLcl

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Questions? Ask here:



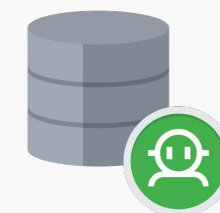
Display username & connection name

```
DEMO@dev_demo > select 1 from dual;
```

```
1
```

```
-----  
1
```

```
> set sqlprompt "@|blue _USER|@@@|green _CONNECT_IDENTIFIER|@ > "
```



Display status bar

TERMINAL

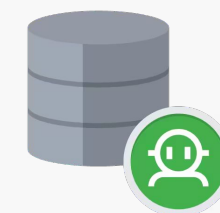
DEMO@dev_demo >

DEMO | dev | NONE | /Users/rg/DEMO/ | None | No time

USERNAME | GIT | PROJECT | CWD | TRANSACTION STATUS | TIMING

> set statusbar on

> set statusbar default username git project cwd txn timing



Customise SQLcl

Set line numbers on | off

```
DEMO@dev_demo > set line numbers on  
DEMO@dev_demo > select 'I really like SQLcl!'  
2* from dual;
```

'IREALLYLIKESQLCL!'

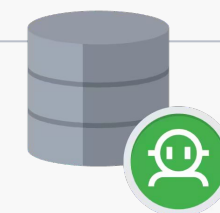
I really like SQLcl!

```
DEMO@dev_demo > set line numbers off  
DEMO@dev_demo > select 'But I dont like line numbers'  
from dual;
```

'BUTIDONTLIKELINENUMBERS'

But I dont like line numbers

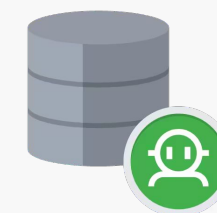
```
DEMO@dev_demo > |
```



Set termout on | off

Controls whether to display the results of SQL when running a script

```
TERMINAL
DEMO@dev_demo > set termout on|
```



Set serveroutput on | off

Print or not output generated by DBMS_OUTPUT

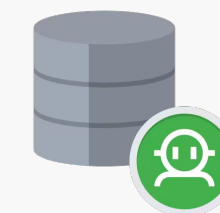
```
DEMO@dev_demo > set serveroutput on
DEMO@dev_demo > exec dbms_output.put_line('Hello!');
Hello!

PL/SQL procedure successfully completed.

DEMO@dev_demo > set serveroutput off
DEMO@dev_demo > exec dbms_output.put_line('No message will be shown');

PL/SQL procedure successfully completed.

DEMO@dev_demo > 
```

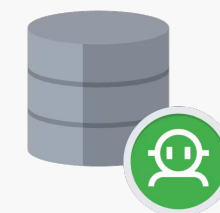


Set feedback on | off

Controls whether to display the number of records returned/affected by a statement

TERMINAL

```
DEMO@dev_demo > set feedback on|
```



Customise SQLcl

Set define on | off | '&' | '#'

Enable / disable prompting

TERMINAL

```
DEMO@dev_demo > sh|
```



Customise SQLcl

Show all params

Display all currently set parameters

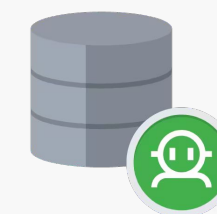
```
DEMO@dev_demo > s|
```



Customise SQLcl

Do I need to set those parameters every time?

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Customise SQLcl

Define login.sql script

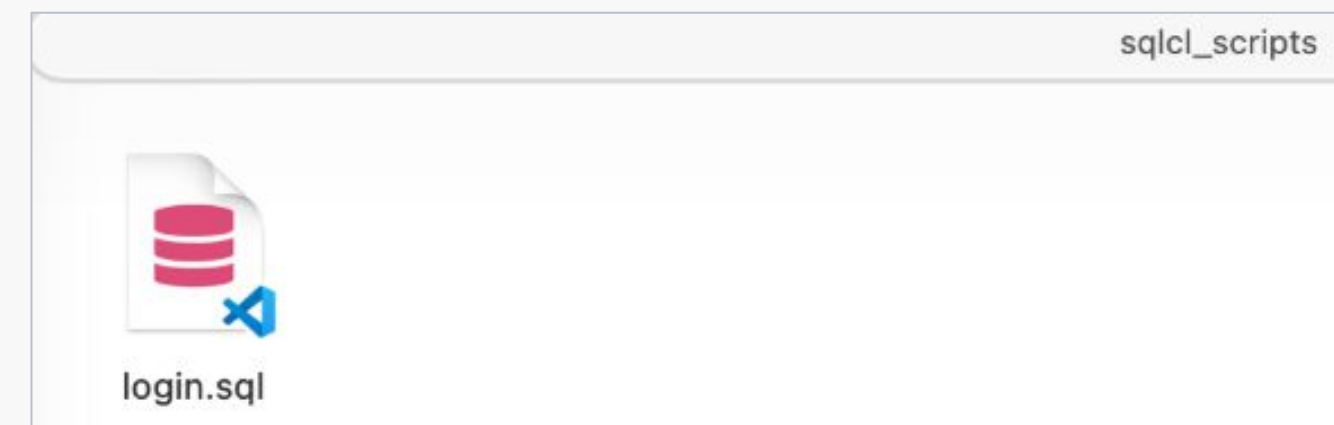
- Add another path into config file -> Users/<your_user>/ .zshrc (or .bash_profile)]

```
export SQLPATH="/Users/rg/apps/sqlcl_scripts/"
```

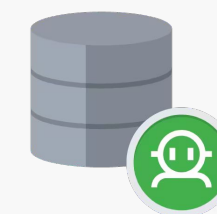


```
.zshrc

export PATH="/Users/rg/apps/liquibase:$PATH"
export PATH="/Users/rg/apps/oracle/sqlcl/bin:$PATH"
export SQLPATH="/Users/rg/apps/sqlcl_scripts/"
export TNS_ADMIN="/Users/rg/apps/oracle/"
```



* script will be executed at each SQLcl start



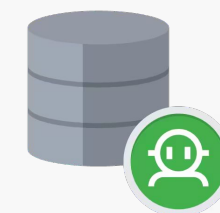
Customise SQLcl

Define custom settings only once

 login.sql 1 X

Users > rg > Downloads >  login.sql > ...

```
1  set sqlprompt "@|blue _USER|@@|green _CONNECT_IDENTIFIER|@ > "  
2  set linenumbers off  
3  set define '&  
4  set feedback on  
5  set serveroutput on  
6  set termout on  
7  set sqlformat ansiconsole  
8  set statusbar on  
9  set statusbar default username git project cwd txn timing  
10
```



BONUS Auto-update SKILLS each last weekday of month

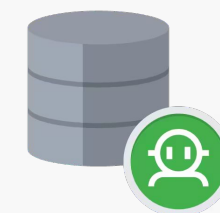
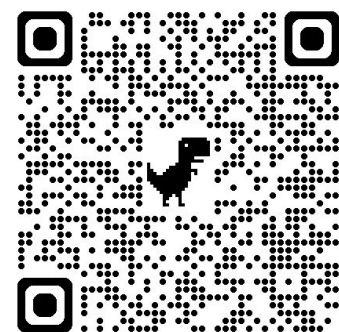
```
o + sqlcl_demo sql -name demo
SQLcl: Release 26.1 Production on Wed Jun 24 09:42:07 2026

Copyright (c) 1982, 2026, Oracle. All rights reserved.

SUCCESS: Synchronized DBTools skills repository to /Users/rg/.dbtools/skills/repositories/oracle-skills version latest (refs/heads/main -> b31c5175d73045297f5d44658dffc03fe4659d8c).
SUCCESS: Installed 0 entries into agent skill locations.
WARN: Skipped existing skill entries: /Users/rg/.agents/skills/apex, /Users/rg/.agents/skills/apex, /Users/rg/.agents/skills/apex, /Users/rg/.agents/skills/db, /Users/rg/.agents/skills/db, /Users/rg/.agents/skills/db, /Users/rg/.agents/skills/fusion, /Users/rg/.agents/skills/fusion, /Users/rg/.agents/skills/fusion, /Users/rg/.agents/skills/graal, /Users/rg/.agents/skills/graal, /Users/rg/.agents/skills/graal, /Users/rg/.agents/skills/oci, /Users/rg/.agents/skills/oci, /Users/rg/.agents/skills/oci, /Users/rg/.copilot/skills/apex, /Users/rg/.copilot/skills/db, /Users/rg/.copilot/skills/fusion, /Users/rg/.copilot/skills/graal, /Users/rg/.copilot/skills/oci
SUCCESS: DBTools skills sync log: /Users/rg/.dbtools/skills/logs/skills-sync-20260624T074212.521Z.log
Connected to:
Oracle AI Database 26ai Standard Edition 2 Release 23.26.2.0.0 - Production
Version 23.26.2.0.0

DEMO_RAFAŁ@demo >
```

More in my
blog post [here](#):



Customise SQLcl



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Q&A



Audience Q&A



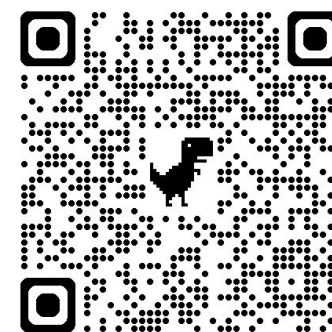


Oracle SQLcl: What Every APEX & PL/SQL Dev Should Know (DBAs, You're Invited too!)

Updated June 8, 2026 • 3 min read



My blog post article
– hub for all things SQLcl [here:](#)



“Migrating Legacy Oracle the Smart Way - OmniSense, Forms2X & the Path to Oracle APEX”

TOMORROW at POUG 2026

Saturday, Sep 05, 2026 at 09:55 AM - 10:40 AM

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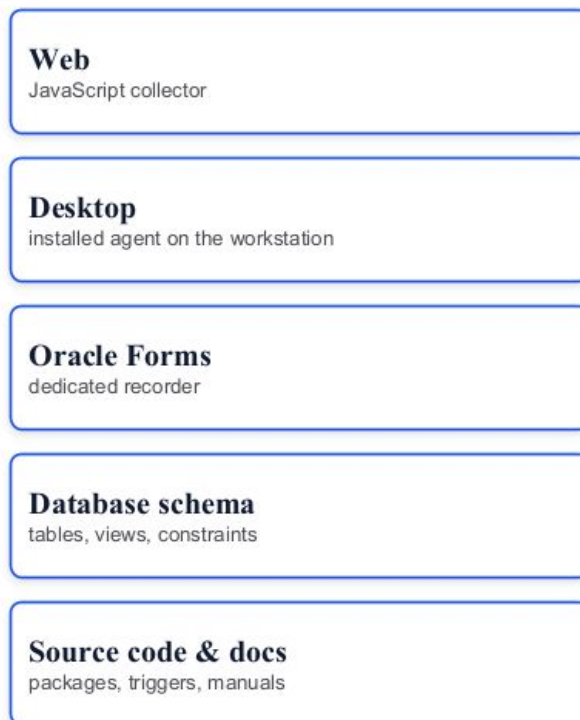


Bartłomiej Jajkowski
Delivery Manager

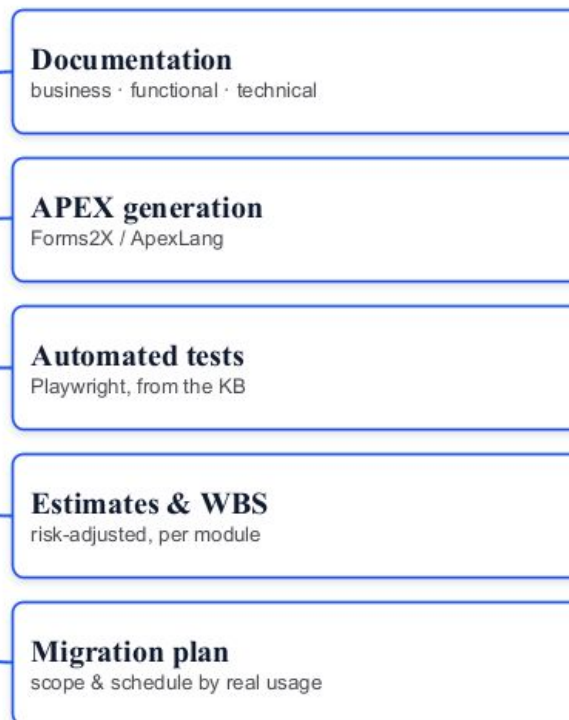
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Our solution

COLLECTORS



OUTPUTS



This focus on the right tools is why we also use **SQLcl** every day.



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for public content related to my
presentations at various conferences**

github.com

<https://github.com/rgrzegorzcyk/conferences>

Thank you for attending!



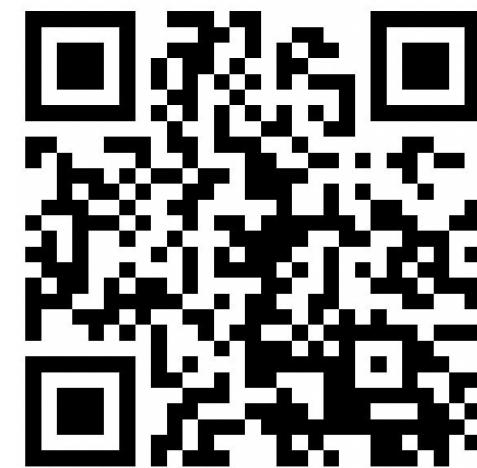
Rafał Grzegorzczak

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[linkedin.com/in/rgrzegorzczak](https://www.linkedin.com/in/rgrzegorzczak)

<https://rafal.hashnode.dev>



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